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AIDC Coverage Universe

The 40 companies under coverage, one entry each — position, evidence, falsifier and recommended strategy

Prepared 16 August 2026 · **Coverage** 40 companies, one Profiler dossier each, researched 7–10 August 2026 · **Company facts** No new research — every citation is a source already catalogued in a dossier
Policy The FEOC, tariff and FCC sections were re-verified against current sources on 15–16 August 2026 and supersede the prior edition · **Classification** Internal — investor education

PRESENTATION STYLE — Axios Smart Brevity. Built to be skimmed; the full record lives in the tables and in the source Markdown.

HOW TO READ THIS COMPANION

This is the reference half of the **AIDC Market Report**. The report carries the argument; this document carries the 40 companies the argument is about, one entry each. Nothing here is summarised away — each entry gives the company’s position, the published evidence for it, what would falsify it, and a **recommended strategy** with the reasoning behind it.

It is built to be looked things up in rather than read front to back, and it is reissued on a different clock from the report: company facts move on earnings, roughly quarterly, while the report’s argument moves more slowly. Terms used here are defined in the report’s primer; the citation and ANALYSIS conventions are identical.

Contents

1. Buyers, Builders and Landlords

The previous chapters described the market as a set of forces: power scarcity, equipment lead times, an architecture reset, a labor shortage. This chapter turns the same market around and looks at it as a set of *companies* — what each one actually is in the pipeline, what is genuinely acting on it this quarter, and what we think it should commercially do about it. It is written to be returned to one entry at a time.

1.0 How to read this chapter

There is one lens that explains more of this sector than any other, and almost nobody states it plainly. In every data-center megawatt, three different companies make three different decisions, and they are rarely the same company:

WHO SPECIFIES, WHO PAYS, WHO INSTALLS

The specifier decides what the equipment must be. It writes the architecture, the reference design, the qualification list. It may never buy a single unit. NVIDIA is the purest case: it sells no power equipment at all, yet its 800 VDC partner slate is the vendor list for the 2027 generation (section 5).

The payer signs the cheque and carries the asset or the offtake. The hyperscalers are payers — but most of them do not choose the equipment brand. They choose a *counterparty*, and the counterparty chooses the brand.

The installer converts equipment into energised megawatts using the scarcest input in the sector, trained craft labor. Installers set the real delivery date, and increasingly they set the product too.

A fourth role sits outside the fence: **the electron seller** — the generator or utility that supplies the power and, by doing so, sets its price. Constellation Energy is the price-setter for firm clean power in this cycle.

Once you hold those roles apart, a company's environment stops looking like a list of headlines and starts looking mechanical. A payer that buys through counterparties is exposed to counterparty behaviour, not to equipment prices. A specifier's leverage is entirely in the qualification calendar. An installer's constraint is headcount and the four-year apprenticeship behind it, so its only genuine growth lever is moving work into a factory. A landlord's economics live or die on how standardised it can make a megawatt. Where a company plays two roles at once — and several here do — it captures margin the single-role players cannot, and it takes on risk the single-role players avoid.

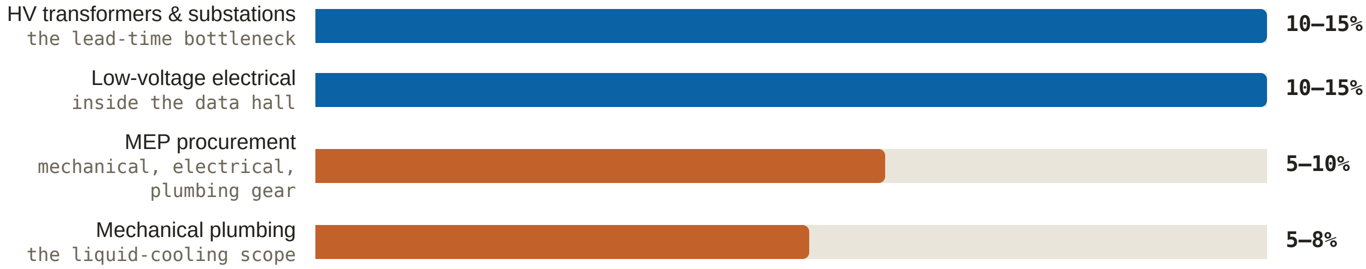
Two structural facts frame everything below. First, *announced* capacity and *energised* capacity are different numbers by roughly an order of magnitude across this cohort. That gap is where the risk and the opportunity both sit.

≈0.3 GW	1.0 GW	1 GW+	≈4.9 GW
Stargate capacity operational, against >9 GW projected by 2029	xAI nameplate compute draw — from zero in 2023	CoreWeave active power, against 3.5+ GW contracted	Crusoe contracted, against a claimed 40+ GW pipeline

Source: Epoch AI Stargate site tracking — OpenAI dossier; SpaceX Form S-1, 2026-05-20 — xAI dossier; Q1 2026 results, 2026-05-07 — CoreWeave dossier; Crusoe contracted-capacity disclosure, 2026-07 — Crusoe dossier.

Second, the money inside a campus does not land where a layperson expects. It lands overwhelmingly on labor and on the electrical gear that labor installs. Quanta Services publishes the only clean public arithmetic on this, and it is the most useful sizing tool in the sector: roughly **$13.5M of craft-led spend per megawatt**, excluding everything outside the fence *(2026 Investor Day, 2026-03-31 — Quanta dossier)*. Within that, the slices break down as follows.

Figure 13: Where the money lands inside the fence (share of a load-center budget, %)



Source: Quanta Services 2026 Investor Day, 2026-03-31 — Quanta dossier. Bars are drawn at the midpoint of each disclosed range; the largest midpoint is set to full width. Quanta claims balance-of-plant coverage of 80–90% of a data-center build, so these slices are close to the whole addressable inside-the-fence budget.

Read the figure this way: the two biggest slices are precisely the two with the worst supply position in the market — grid gear on multi-year lead times (section 3) and electrical work performed by a labor pool that takes four years to train (section 7). That is not coincidence. It is the reason schedule, not price, is the currency of this sector.

1.1 NVIDIA — the specifier

WHAT THEY ARE

The company that sells the computers, and therefore the company that decides what every power vendor must build — without buying any power equipment itself.

THE ENVIRONMENT

NVIDIA's problem is no longer demand. FY2026 revenue was $215.9B (+65%), data center $193.7B, and its own forward supply commitments nearly doubled in a single quarter from $50.3B to $95.2B *(Q4/FY2026 results, 2026-02-25 — NVIDIA dossier)*; Q1 FY2027 ran $81.6B (+85%) with data center at $75.2B and Q2 guided to $91B *(Q1 FY2027 results, 2026-05-20 — NVIDIA dossier)*. A supply commitment is a non-cancellable purchase obligation NVIDIA has taken on with its own suppliers — so a near-doubling is the most honest forward-demand signal the company emits, because it costs real money if demand disappoints.

Three forces are actually acting on it. **First, concentration.** Four direct customers were 61% of Q3 FY2026 revenue *(Q3 FY2026 10-Q, 2025-11-19 — NVIDIA dossier)*. **Second, circularity.** NVIDIA has committed to invest up to $100B in OpenAI as OpenAI deploys NVIDIA systems *(OpenAI 10 GW partnership, 2025-09-22 — NVIDIA dossier)* and up to $10B alongside Anthropic's initial commitment of up to 1 GW *(Anthropic/Microsoft partnership, 2025-11-18 — NVIDIA dossier)*. When a supplier funds its customer's purchases, revenue growth and financing capacity become the same variable. **Third, China is now downside only** — guidance assumes zero China data-center compute revenue, against a potential ≈$2.9B SAMR fine *(CNBC, 2025-09-15 — NVIDIA dossier)*.

The lever that matters for this report is the architecture. NVIDIA's 800 VDC design moves ≈157% more power through the same copper versus 415 VAC and cuts up to 45% of it, and it explicitly prescribes storage in two places — supercapacitors at the rack for millisecond transients, facility-level batteries at the interconnection for second-to-minute smoothing *(800 VDC whitepaper, 2025-10-13 — NVIDIA dossier)*. Kyber racks arrive at ≈600 kW each in 2H 2027 *(Tom's Hardware, 2025-03-18 — NVIDIA dossier)*. ANALYSIS – MODERATE GB300's own rack-level smoothing (≈65 J/GPU, up to −30% peak grid demand) internalises some battery revenue but legitimises rack storage faster than it cannibalises it, with the AI-server BBU market estimated at ≈$2.8B in 2026 rising to ≈$7.1B by 2033 *(TrendForce — NVIDIA dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Keep giving the power architecture away. NVIDIA's power specification costs it almost nothing and buys it two things no competitor can match: it makes the entire electrical supply chain plan its factories around NVIDIA's rack roadmap, and it converts a physical constraint — copper, transformers, floor space — into a reason to buy the newest generation rather than the cheapest. Publish more of it, earlier, and widen the named-partner slate deliberately to include the grid tier, which it already began doing at OCP. The second imperative is to break the concentration optics: every gigawatt of demand that arrives from a customer NVIDIA has not financed is worth more to the equity story than a gigawatt that arrives from one it has, because the circular structure means vendor-financed demand cannot be underwritten twice. Practically, that argues for pushing hard on sovereign, enterprise and non-frontier-lab channels even at lower average selling prices.

WHAT WOULD CHANGE THE VIEW

The supply-commitment line. If forward supply commitments flatten or fall for two consecutive quarters while revenue still grows, NVIDIA has stopped believing its own order book, and every power supplier building 800 VDC capacity for 2027 is building into a softer market than it thinks.

1.2 Microsoft — the payer who buys firmness, not batteries

WHAT THEY ARE

The best-underwritten buyer in the sector, and the one that has deliberately chosen to buy reliable grid power rather than build reliability into its own buildings.

THE ENVIRONMENT

Microsoft is the only hyperscaler whose backlog is growing faster than its spending. FY2026 Q4 revenue was $90.0B (+18%), Azure grew 43% and crossed $100B for the year, and commercial remaining performance obligations reached **$678B, up 84%** *(FY2026 Q4 results, 2026-07-29 — Microsoft dossier)*. Remaining performance obligation, or RPO, is contracted revenue not yet delivered — a signed promise to pay. Against a ≈$190B calendar-2026 capex plan announced versus $154.6B consensus *(CNBC Q3 FY2026, 2026-04-29 — Microsoft dossier)*, later restated to ≈$175B purely through longer useful lives and lease treatment rather than a real cut *(CFO Dive, 2026-07 — Microsoft dossier)*, that backlog is the strongest evidence in the whole cohort that the buildout is pre-sold. ANALYSIS – HIGH The OpenAI $250B incremental Azure commitment alone underwrites years of Fairwater capacity *(Microsoft dossier)*.

The second force is architectural and it is the most commercially consequential fact about Microsoft in this report. Fairwater Atlanta runs on resilient grid power with **no on-site generators and no UPS** — approximately 140 kW racks and 1,360 kW rows, delivering four-nines availability at three-nines cost *(AI superfactory architecture, 2025-11-12 — Microsoft dossier)*. Removing the uninterruptible power supply and the diesels removes capital cost, maintenance, floor space and a failure mode; it also removes the single largest battery purchase in a conventional data center. ANALYSIS – HIGH Microsoft has structurally chosen grid procurement over on-site storage, which makes its utilities and grid partners the channel and leaves the pledged worldwide replication of its Dublin grid-interactive UPS pilot the one door still ajar *(Microsoft dossier)*.

Third, its firmness is nuclear-weighted and mostly not yet delivered: the ≈835 MW Three Mile Island restart as the Crane Clean Energy Center, accelerating toward 2027 behind a $1B federal loan, an NRC waiver and FERC clearance *(Utility Dive, 2026-06-02 — Microsoft dossier)*; the world's first fusion power purchase agreement with Helion at ≥50 MW targeting 2028 *(Helion, 2023-05-10 — Microsoft dossier)*; and a 10.5+ GW Brookfield renewables framework *(Brookfield, 2024-05-01 — Microsoft dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Microsoft should keep buying grid firmness and keep refusing to own generation — but it should start paying for *optionality on flexibility* rather than more capacity. The no-UPS design is correct on cost and wrong on one risk: it makes Microsoft wholly dependent on a grid whose large-load tariffs are being rewritten in public, in front of angry ratepayers. The cheapest hedge is not batteries in the building; it is contracted demand flexibility, which is what its own Dublin grid-interactive concept always was. Microsoft can offer utilities curtailability — a contractual commitment to cut its own load on request, which is what the grid calls curtailment — in exchange for interconnection priority, exactly the trade Google has already been signing. Second, it should lean harder on the one asset no rival can copy: an $678B backlog is collateral. Using contracted revenue to underwrite third-party generation and transmission — rather than putting the asset on its own balance sheet — extends the Crane template at the lowest possible cost of capital, and keeps Microsoft's capex line defensible in quarters when Azure's growth rate wobbles, which is the only number its equity actually trades on.

WHAT WOULD CHANGE THE VIEW

A Crane slip past 2027. On-time delivery validates buying firmness upstream; a slip pushes Microsoft's own firming demand toward gas and storage, and would make the no-UPS design look like an unhedged bet rather than a cost win.

1.3 Google — the payer that became a builder

WHAT THEY ARE

The only hyperscaler that bought the company that develops its power, and therefore the only one that specifies, pays for and owns its own generation and storage.

THE ENVIRONMENT

Google is spending at the top of the cohort and financing it externally. 2026 capex was guided to $195–205B after a third raise in one year, Cloud grew 82% to $24.8B with a $514B backlog, and quarterly free cash flow was **−$5.9B** *(Alphabet Q2 2026 8-K, 2026-07-22 — Google dossier)* against a first $400B revenue year *(Alphabet Q4/FY2025 8-K, 2026-02-04 — Google dossier)*. ANALYSIS – MODERATE Funding has shifted decisively to capital markets, with roughly $95B of debt and equity raised across late 2025 and 2026 while free cash flow is negative — sustainable at Alphabet's earnings power, but it makes the capex programme hostage to equity-market patience in a way it was not in 2024 *(Google dossier)*.

What it bought with that money is the interesting part. Google acquired renewables developer Intersect Power outright for ≈$4.75B, bringing ≈2.2 GW of solar and 2.4 GWh of batteries in-house along with the "energy park" model — data centers co-located with behind-the-meter solar and storage specifically to avoid the interconnection queue, first at Haskell County, Texas, with 640 MW of solar and 1.3 GWh of batteries *(PV Tech, 2026-03 — Google* *dossier)*. It anchored the largest battery project by energy capacity announced anywhere — 300 MW / **30 GWh** of Form Energy iron-air storage via Xcel serving Pine Island, Minnesota *(Google, 2026-02-24 — Google dossier)* — and the largest US solar-plus-storage power purchase agreement at Steel River, building to 2.5 GW of solar and 2.9 GWh of batteries *(pv magazine, 2026-07-15 — Google dossier)*. It also signed 1 GW of data-center demand response across five utilities *(Google, 2026-03-19 — Google dossier)*. Iron-air at 100-hour duration is a different product class from the four-hour lithium systems that dominate the market: it is not a peak-shaving asset, it is a substitute for firm capacity, which is why it is the deal that most changes what "storage" means to a hyperscaler.

The pressure behind all of it is that its own clean-energy goal is losing. Hourly carbon-free matching sits at 66% while total footprint is up 81% versus the 2019 baseline *(2026 Environmental Report, 2026-06 — Google dossier)*. ANALYSIS – MODERATE The interconnection queue is the strategic enemy Google is engineering around, and the playbook rivals will copy *(Google dossier)*. Separately, its TPU programme has crossed into merchant silicon, with Anthropic anchored at up to ≈1M TPUs and Meta signed as a multi-year lessee *(Google dossier)* — ANALYSIS – HIGH the only at-scale alternative to NVIDIA among hyperscalers *(Google dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Google should productise the energy park and sell it. It now owns the rarest combination in the sector — a development arm, a load to anchor it, and a balance sheet — and the marginal value of the tenth park it builds for itself is lower than the value of being the counterparty other people's data centers plug into. Concretely: keep Intersect building, but sell capacity and firmness from those parks to colocation operators and neoclouds who cannot self-develop, using the Pine Island structure where Google pays the full incremental cost through a dedicated tariff. That structure is the single most politically durable thing Google has done, because it answers the ratepayer objection before it is raised. Second, spend on duration rather than more four-hour lithium. The 30 GWh iron-air anchor already gives Google a first-mover position in the only storage class that substitutes for a gas plant; a second and third long-duration anchor would let it retire the "moonshot getting harder" narrative with physics instead of accounting.

WHAT WOULD CHANGE THE VIEW

Whether a second energy park reaches commercial operation on schedule after Haskell. The thesis is that owning development compresses time-to-power; one park proves nothing, two on schedule proves the model, and a slip proves that owning a developer buys you the developer's problems rather than solving them.

1.4 Amazon — the payer who buys through other people

WHAT THEY ARE

The largest single AI-infrastructure programme and the largest corporate clean-energy buyer on earth — which almost never picks the equipment brand itself.

THE ENVIRONMENT

The programme re-accelerated and got more expensive at the same time. Q2 2026 was Amazon's first $200B+ revenue quarter, AWS grew **36.7%** to $42.2B — its fastest in eighteen quarters — 2026 capex guidance rose from ≈$200B to ≈$220B, net income of $62.6B included a $53.4B mark-up on its Anthropic stake, and quarterly free cash flow was −$7.6B *(Q2 2026 earnings, 2026-07-30 — Amazon dossier)*. The AWS backlog was $244B, up 40%, at FY2025 *(FY2025 8-K, 2026-02-05 — Amazon dossier)*. Most importantly for this report, Amazon put gigawatts into an earnings filing: Anthropic up to **5 GW** of Trainium capacity, OpenAI ≈2 GW, and 2.1M+ AI chips deployed in twelve months *(Q1 2026 8-K, 2026-04-29 — Amazon dossier)*. When a filing denominates compute in gigawatts, compute procurement has formally become power procurement.

The energy machine is the deepest in the cohort — 40+ GW across 700+ projects including eleven utility-scale battery projects *(carbon-free energy leadership post — Amazon dossier)*, flagship among them Bellefield with AES at 1 GW of solar plus 1 GW / 4,000 MWh of storage at completion, phase one of 500 MW and 1 GWh finished in June 2025 *(Energy-Storage.News, 2025-06-12 — Amazon dossier)*. But the buying centre is not Amazon. ANALYSIS – HIGH Amazon's storage procurement splits into three layers that must not be conflated: utility-scale batteries where Amazon signs the offtake and the *developer* owns the asset and chooses the cells, so the equipment decision sits with AES and Primergy; behind-the-meter batteries at the campus, where there are no announced Amazon deployments at all; and rack-level backup inside the white space, which is where AI-specific battery demand actually lands *(Amazon dossier)*. Reporting placing Samsung SDI in that third layer via Taiwan's Simplo remains unconfirmed by either company ANALYSIS – LOW *(UPI, 2026-04-28; Digitimes, 2026-07-07 — Amazon dossier)*.

Third, Amazon wrote the rulebook everyone else now works under. Its behind-the-meter nuclear arrangement with Talen was rejected by FERC and the restructured front-of-meter contract, ramping to 1,920 MW of Susquehanna output through 2042, became the template *(Utility Dive, 2024-11; Talen, 2025-06-11 — Amazon dossier)*. ANALYSIS – MODERATE That precedent pushes incremental firming demand toward grid-side assets, including storage *(Amazon dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Amazon should stop outsourcing the equipment decision in the one layer where it has no substitute. Layers one and three are working: developer-owned utility-scale batteries are the cheapest way to buy gigawatt-hours, and rack-level backup arrives with the servers. Layer two — campus-level storage — is empty, and that is a genuine exposure rather than an elegant choice, because the fourteen-hour us-east-1 failure that affected roughly 70,000 organisations *(AWS post-event summary, 2025-10 — Amazon dossier)* demonstrated that Amazon's concentration risk is operational, not just commercial. A campus-level medium-voltage storage standard, specified once and deployed as a repeatable block, would give it ride-through it currently rents from utilities and a schedule weapon in markets where interconnection is the gate. Second, Amazon should convert its most under-used asset — the largest corporate renewables portfolio in the world, built over more than a decade — into a merchant firmness product for its own colocation and neocloud customers. It already carries the counterparty relationships; it is currently monetising them only as a cost centre.

WHAT WOULD CHANGE THE VIEW

Free cash flow. Amazon has funded this with its balance sheet on the argument that AWS growth validates it; a quarter where AWS growth decelerates while capex guidance holds would reprice the entire buildout, because the market has been paying for the acceleration, not the assets.

1.5 Meta — the highest-beta payer in the cohort

WHAT THEY ARE

The most aggressive energy buyer relative to its size, and the only member of the big four with no external cloud revenue to absorb what it is spending.

THE ENVIRONMENT

The escalation is the steepest in the sector and the cash cost is now visible. Q2 2026 revenue grew 28% to $60.8B but earnings per share missed on a 55% cost jump; quarterly capex of $31.08B consumed roughly 98% of operating cash flow and free cash flow fell to **$784M**, down 91% year on year *(Meta Q2 2026 results, 2026-07-29; CNBC coverage, 2026-07-29 — Meta dossier)*, against 2026 capex guidance of $130–145B. ANALYSIS – HIGH Meta is the highest-beta energy buyer in the covered set: with no cloud revenue absorbing the spend, its capex and its 6.6 GW nuclear package are bets that superintelligence-era products monetise, and the free-cash-flow collapse is the measurable cost of that bet *(Meta dossier)*.

Its answer has been to move the buildings off its own balance sheet. The Blue Owl joint venture financed Hyperion at ≈$27B on an 80/20 split, and BlackRock did the same for the ≈1 GW El Paso campus at ≈$14B *(Meta, 2025-10-21; Meta, 2026-07-28 — Meta dossier)*. ANALYSIS – MODERATE The joint-venture financing model is now core strategy: it preserves reported capex optics and credit capacity while transferring residual-value risk to private capital, and should be expected on every subsequent gigawatt campus *(Meta dossier)*. Hyperion itself expanded to 5 GW and $50B+, with an Entergy package of seven new gas plants plus three grid-scale batteries *(Meta, 2026-07-13 — Meta dossier)*; Prometheus runs on 200 rising to 400 MW of behind-the-meter gas plus AEP transmission *(Data Center Frontier — Meta dossier)*; and contracted nuclear totals ≈7.7 GW, the largest corporate commitment anywhere *(Meta, 2026-01-09; Meta, 2025-06-03 — Meta dossier)*.

Its storage, by contrast, sits on the utility's side of the meter. The clearest example is Enbridge Cowboy — 365 MW of solar and a 200 MW / 1,600 MWh battery built on Tesla hardware, with the *utility* tolling the battery rather than Meta owning it *(Utility Dive, 2026-05-21 — Meta dossier)* — alongside a reservation of up to 1 GW / 100 GWh of Noon Energy long-duration storage with a 25 MW pilot in 2028 *(Meta, 2026-04-27 — Meta dossier)*. ANALYSIS – MODERATE Meta buys grid firmness as a service, so the channel for storage suppliers is Meta's utility counterparties, not Meta *(Meta dossier)*. The exposure that follows is political: ANALYSIS – MODERATE the Louisiana gas controversy is the sector's regulatory bellwether, with a judge compelling disclosure of withheld demand evidence and a commissioner dissenting over stranded-asset risk on fifteen-year Meta contracts against thirty-year plant lives *(Meta dossier; UCS — Meta dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Meta's exposure is not its spending level, it is the *shape* of its power commitments. It has bought long-lived firm generation — gas plants and reactors with thirty-year lives — against fifteen-year contracts and a product roadmap with no revenue line. That asymmetry is exactly what its regulators are attacking, and it is the reason the Louisiana docket is dangerous out of proportion to its size. The fix is to rebalance the portfolio toward assets whose value survives being wrong: storage and demand flexibility. A battery is re-deployable and re-contractable; a purpose-built combined-cycle plant in Richland Parish is not. Meta should therefore convert a meaningful share of its Entergy firming scope into utility-tolled storage — a structure it has already proven with Enbridge — and it should follow Google in signing contracted curtailability, which costs almost nothing when your workload is training rather than serving. Second, keep industrialising the joint-venture model, but expect the price of private capital to rise as residual-value risk becomes better understood; locking more of it now is cheaper than locking it later.

WHAT WOULD CHANGE THE VIEW

The Louisiana Public Service Commission outcome on Entergy phase two. Approval on Meta's terms confirms that a hyperscaler can still get several gigawatts of purpose-built gas approved; a conditioned or rejected outcome resets how much firm generation gets built for AI load nationally, and pushes that demand toward storage.

1.6 OpenAI — the demand that structures the market

WHAT THEY ARE

Not a builder and barely a payer in cash terms — the entity whose announced commitments set other companies' order books, and increasingly a power buyer in its own right.

THE ENVIRONMENT

The scale of the commitments is the fact. Roughly **$1.4T and ≈30 GW** of infrastructure commitments were publicly acknowledged around the October 2025 restructuring, and the January 2025 target of 10 GW by 2029 had been surpassed *in commitments* by April 2026, with more than 3 GW added in a single trailing ninety-day window *(Reuters factbox, 2025-10; Stargate scaling update, 2026-04 — OpenAI dossier)*; planned compute spend through 2030 was raised to $750B *(WSJ via Yahoo, 2026-07-22 — OpenAI dossier)*. Against that, independent site tracking finds ≈0.3 GW operational at the Abilene flagship *(Epoch AI — OpenAI dossier)*. ANALYSIS — MODERATE Delivery risk is concentrated in the announced-to-operational gap, and every site depends on the same constrained turbines, transformers and grid queues documented elsewhere in this report *(OpenAI dossier)*.

The financing is the second force, and it is structural rather than incidental. ANALYSIS — HIGH The model is circular by construction and this is its central fragility: NVIDIA invests up to $100B as OpenAI deploys NVIDIA systems, Amazon invests while selling cloud, Oracle borrows against OpenAI receivables, and Microsoft's equity-method losses imply roughly $11.5B of quarterly OpenAI losses — the same dollars appearing as vendor revenue, OpenAI funding and counterparty backlog at once *(OpenAI dossier)*. A $122B round closed at $852B post-money in March 2026 and a confidential S-1 followed in June *(OpenAI, 2026-03-31; S-1 filing, 2026-06-08 — OpenAI dossier)*; reported revenue was ≈$25B annualised against roughly $3.7B of quarterly cash burn *(OpenAI dossier)*.

Third — and most under-appreciated — OpenAI has begun buying power directly. The SB Energy partnership contracts generation-side development for multi-gigawatt campuses including a 1.2 GW Texas facility, rather than going through a cloud landlord *(Stargate SB Energy partnership, 2026-01-09 — OpenAI dossier)*. Its own policy filing recommends a US target of 100 GW a year of new capacity, cites an "electron gap" of 429 GW added in China in 2024 against 51 GW in the US, asks for interconnection fast-tracking for storage, and discloses that its data centers are designed to be curtailable *(OSTP RFI filing, 2025-10-27 — OpenAI dossier)*. Its Stargate sites span the entire behind-the-meter menu: gas microgrids at Shackelford, gas plus fuel cells at Doña Ana, roughly 70% renewables-and-battery at Port Washington, and utility grid plus batteries at Saline Township *(Epoch AI — OpenAI dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* OpenAI should stop announcing gigawatts and start announcing energisations. Its announcement cadence has been an extraordinarily effective financing instrument — it moves supplier order books and share prices — but the instrument is depreciating, because the market has now learned to discount announced capacity against the 0.3 GW that is actually running. The highest-value disclosure OpenAI could make is a quarterly operational-megawatt figure, the way xAI's filing does with nameplate compute draw. Second, lean into the curtailability it has already disclosed and make it contractual. OpenAI's workload is uniquely flexible: training can move in time and across sites, which is the single most valuable thing a grid operator can be offered right now. Selling flexibility buys interconnection priority and community goodwill at close to zero marginal cost, and it is the only lever OpenAI holds that does not require raising more money. Third, keep buying generation development directly — the SB Energy structure removes a landlord's margin and, more importantly, removes a landlord's schedule from OpenAI's critical path.

WHAT WOULD CHANGE THE VIEW

Whether Abilene reaches its projected 1.2 GW by the end of 2026. It is the cleanest single test of whether the announced-to-operational gap is a timing artefact or a structural ceiling, and every counterparty backlog in this chapter is priced on the answer.

1.7 xAI — the self-builder

WHAT THEY ARE

The company that proved a gigawatt AI campus can be built and powered without waiting for the grid — by buying its own power plant and its own batteries.

THE ENVIRONMENT

The build record is the asset. Colossus reached **≈1.0 GW of nameplate compute draw** by 31 March 2026 from zero in 2023, powered almost entirely by self-deployed behind-the-meter generation, with the filing confirming the campus is capable of operating entirely off its own plant — the grid is the supplement, not the source *(SpaceX Form S-1, 2026-05-20 — xAI dossier)*. Independent verification puts ≈460 MW of on-site gas across the two campuses and clocks cluster bring-ups at 91 and 64 days *(SemiAnalysis, 2025-12 — xAI dossier)*. Buffering it is the largest Megapack deployment in the world, with more than 240 units at Colossus 1 alone, purchased at $191M in 2024, $506M in 2025 and $34M in Q1 2026 *(SpaceX Form S-1, 2026-05-20 — xAI dossier)*. Batteries are what make this work: a gas turbine cannot follow a GPU cluster's second-to-second swings, so the battery absorbs the transient and the turbine runs steady. Without the storage layer, off-grid operation at this density is not physically available.

The second force is procurement velocity, which distorts the market it operates in: ≈$3.7B of turbine commitments in four months of 2026, comprising $805M and $925M purchase agreements running through 2029 plus a ≈$2.0B mobile-fleet acquisition *(SpaceX Form S-1, 2026-05-20 — xAI dossier)*. ANALYSIS – HIGH Speed-to-power, not model quality, is the differentiator, and grid allocations of ≈300 MW cover only a fraction of campus load *(xAI dossier)*.

The third force is the bill. The AI segment lost **$6.4B** from operations on ≈$3.2B of revenue in 2025, with a further $2.47B operating loss in Q1 2026 alone *(SpaceX Form S-1, 2026-05-20; TechCrunch, 2026-05-20 — xAI dossier)*. Anthropic's cloud agreements at $1.25B a month through May 2029 — roughly $15B a year — transform that, but carry a ninety-day termination right *(xAI, 2026-05 — xAI dossier)*. And the permitting record is the sector's cautionary template: a federal Clean Air Act suit brought by the NAACP and SELC over allegedly unpermitted turbines, a **$399M** self-recorded litigation accrual, and 69 temporary mobile turbines scheduled out between August 2026 and July 2027 under a state agreed order while a permitted 1.2 GW, 41-turbine permanent plant is built *(CNBC, 2026-04-14; CNBC, 2026-03-10 — xAI dossier)*. ANALYSIS – MODERATE Related-party procurement is structural, and vendors should assume Tesla gets first look at any storage scope *(xAI dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* xAI should convert its build capability into a product and stop treating it as an internal cost. It has demonstrably the fastest energised-megawatt delivery in the industry and, in the Anthropic contract, proof that a rival will pay $15B a year to rent it. That is a merchant power-and-compute business, not a research lab's overhead — and it should be capitalised as one, with the operational metric it already discloses (nameplate compute draw) as the headline. Second, and urgently, it should pay for permitting the way it pays for turbines. The $399M accrual is not the cost of the litigation; it is the price of the schedule it bought, and it will rise. Front-loading full major-source permits, community benefit agreements and emissions controls is cheaper per megawatt than the current approach, and it protects the one thing the strategy actually depends on — the ability to keep doing this in the next county. Third, the related-party storage default is a commercial mistake. Competitively tendering the storage scope at Colossus scale would cut cost per megawatt-hour materially in a market with several credible gigawatt-scale suppliers, and would remove a governance discount that a public listing will price.

WHAT WOULD CHANGE THE VIEW

The mobile-turbine transition. If the permanent 1.2 GW plant is energised and the 69 mobile units leave on the agreed schedule without a load reduction, the self-build template survives contact with regulators. If load has to be cut to meet the removal order, the template's real constraint is legal rather than technical — and every developer copying it is exposed.

1.8 Oracle — the landlord who publishes its power menu

WHAT THEY ARE

The largest external AI-compute landlord on earth: it buys the buildings and the power, rents out the computers, and finances the gap with debt.

THE ENVIRONMENT

The backlog and the cash flow moved in opposite directions, violently. FY2026 remaining performance obligations reached **$638B, up 363%**, on revenue of $67.4B (+17%), with free cash flow of **−$23.7B**, capex of $55.7B, $43B of debt plus $5B of equity raised and roughly $40B more planned including a $20B at-the-market equity programme *(Q4 FY2026 earnings release, 2026-06-10 — Oracle dossier)*. Press reporting attributes roughly half that backlog to a single ≈$300B five-year OpenAI contract beginning in 2027 *(Yahoo/analysts, 2025-12 — Oracle dossier)*. ANALYSIS – MODERATE The RPO overstates firm demand: $75B comes from GPU-prepayment or customer-supplied-GPU structures, the Abilene expansion to 2.1 GW was reversed, and ≈0.3 GW is operational against more than 9 GW projected by 2029 *(Oracle dossier)*. ANALYSIS – HIGH The buildout is structurally debt-and-equity funded with thin near-term economics, with S&P at BBB− and a forecast FY2027 free-cash-flow deficit of ≈$42B *(Oracle dossier)* — and leaked internals put GPU-rental gross margins at ≈14–16% against ≈70% for the legacy stack *(CNBC, 2025-10-07 — Oracle dossier)*.

The second force is what makes Oracle the most useful single account in this report: it publishes how each campus is powered, and each is different. Shackelford County runs fully off-grid on reciprocating engines with 115 MW live before delivery; Project Jupiter at Doña Ana was awarded up to **2.45 GW** of Bloom fuel cells, explicitly displacing previously planned gas turbines and diesel generators; Abilene runs on the ERCOT grid plus roughly 360 MW of on-site turbines; Saline Township pairs DTE grid supply with roughly $300M of new battery storage; Port Washington is around 70% wind, solar and battery *(Project Jupiter announcement, 2026-04-27; Oracle data-center hub — Oracle dossier)*. ANALYSIS – HIGH That per-site diversity is deliberate procurement strategy, not improvisation, and it is the best public map of behind-the-fence options in the sector *(Oracle dossier)*. Codifying it, the January 2026 "we pay our own way on energy" post commits to on-site generation or Oracle-funded grid upgrades with no ratepayer pass-through *(Oracle, 2026-01-26 — Oracle dossier)* — a de facto procurement specification for anyone bidding an Oracle campus.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Oracle should standardise the menu. Running five simultaneously different power architectures was the right way to learn the market and is now the wrong way to build it: every distinct architecture multiplies engineering, spares, commissioning labor and financing diligence, and Oracle is paying that overhead at

14–16% gross margin. Picking two reference architectures — one grid-plus-storage, one fully off-grid generation-plus-storage — and repeating them would compress both schedule and cost of capital, because lenders price repeatable assets more cheaply than bespoke ones. Second, Oracle should sell the power itself, not just the compute. It is already funding generation and grid upgrades on its own balance sheet; contracting surplus firm capacity to co-located tenants converts an energy cost centre into a revenue line at a far better margin than GPU rental, and it uses the one skill the market has not priced — the CFO arrived from Schneider Electric with an AES energy background *(Oracle dossier)*. Third, disclose the customer concentration. The market is already assuming the worst; a contracted-revenue breakdown by counterparty would cost Oracle less than the credit spread it currently pays for the ambiguity.

WHAT WOULD CHANGE THE VIEW

The FY2027 free-cash-flow deficit against the ≈$42B forecast, once OpenAI revenue actually begins contributing. That is the quarter the collectability question converts from theoretical to observable — and Oracle's credit spread is the sector's single best real-time barometer of AI-capex risk.

1.9 CoreWeave — the leveraged tenant

WHAT THEY ARE

The specialist AI cloud that rents its buildings and its power from other people and borrows against its chips to grow.

THE ENVIRONMENT

Growth and losses are both compounding. Q1 2026 revenue was $2.08B (+112%) with a net loss of $740M, total debt of $24.9B, and a backlog that jumped to **$99.4B, up 284%**, alongside the milestone of more than 1 GW of active power across 49 data centers against 3.5+ GW contracted *(Q1 2026 results, 2026-05-07 — CoreWeave dossier)*. FY2025 revenue was $5.13B (+168%) with a $1.17B net loss and interest expense of $1.23B — roughly 24% of revenue — and Microsoft at ≈67% of the total *(FY2025 10-K, 2026-02 — CoreWeave dossier)*. Diversification is real: Meta expanded to $21B, OpenAI to ≈$22.4B, and Jane Street signed $6B *(CoreWeave, 2026-04-09; 2025-09-25; 2026-04-15 — CoreWeave dossier)*.

The second force is a genuine validation the equity market has ignored. The cost of GPU-collateralised debt fell from roughly 15% in 2023 to investment-grade pricing at SOFR plus 225 basis points on the $8.5B DDTL 4.0, rated A3/A-low *(globaldatacenterhub, 2026-03-15 — CoreWeave dossier)*. Sophisticated lenders have repriced the risk of lending against graphics processors by roughly two thirds in three years — that is a considered judgment about residual values, and it is the strongest available counterweight to the short thesis. ANALYSIS – HIGH Even so, the backlog's equity value hinges on renewal economics and depreciation truth rather than the headline: a five-year weighted duration against a six-year useful-life assumption that critics argue should be three to four and three-quarter years, in a market where prior-generation rental rates have fallen 50–70% *(CoreWeave dossier)*.

Third — the structural weakness — CoreWeave does not own its power. ANALYSIS – MODERATE Power-ownership incompleteness is the weakness the failed Core Scientific merger locked in: heavy lease dependence of roughly $3.5B of lease liabilities, a guidance cut in Q3 2025 caused by a single third-party developer running late, and a terminated deal that would have brought ≈1.3 GW in-house *(CoreWeave dossier; Core Scientific termination, 2025-10-30; CNBC, 2025-11-10 — CoreWeave dossier)*. The correction has begun: three Indonesian facilities totalling 360 MW of contracted IT power, online expected 2028, all CoreWeave-owned *(Indonesia expansion, 2026-08-04 — CoreWeave dossier)*, inside an NVIDIA partnership targeting more than 5 GW of AI factories by 2030 *(NVIDIA partnership, 2026-01-26 — CoreWeave dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* CoreWeave should buy power capability, not power assets. Owning gigawatts outright is the wrong use of a balance sheet already carrying $24.9B of debt at nearly a quarter of revenue in interest — but the Q3 2025 guidance cut proved that a leased model transmits *someone else's* schedule directly into CoreWeave's guidance, and no amount of operational excellence fixes that. The intermediate structure is the right one: take equity in the shells and the substations, contract the on-site generation and storage itself, and leave the real estate to landlords. That converts a single point of failure into a diversified one at a fraction of the capital. Second, its genuine moat is operational — the only Platinum-tier rating in SemiAnalysis's cloud assessment, and consistent first-to-new-silicon delivery *(SemiAnalysis ClusterMAX, 2025-03 — CoreWeave dossier)*. It should price that. Being first to Vera Rubin and running 130 kW racks reliably is worth a premium that a commodity-rental framing forfeits. Third, it should settle the depreciation argument with disclosure rather than rhetoric: publishing realised rental rates by GPU generation would either kill the bear case or confirm it, and the ambiguity is currently costing more than the answer would.

WHAT WOULD CHANGE THE VIEW

Microsoft's share of revenue. The anchor tenant has already declined an expansion and routed capacity elsewhere; if Microsoft's proportion falls because Microsoft shrank rather than because others grew, the backlog's renewal assumptions are wrong at the most sensitive point.

1.10 Crusoe — the energy-first developer

WHAT THEY ARE

The developer that builds AI campuses power-first — securing generation and storage before the building — and hands over energised megawatts.

THE ENVIRONMENT

The execution record is exceptional and independently visible. The Stargate Abilene flagship had its first phase live within a year of construction start *(Crusoe, 2025-09-30 — Crusoe dossier)*, and a second Abilene campus of 900 MW for Microsoft pairs two 336 MW critical-IT-load buildings with a 900 MW on-site plant *plus* medium-voltage battery storage, first building mid-2027 *(Crusoe, 2026-03-27 — Crusoe dossier)*. ANALYSIS – HIGH Time-to-power execution at scale is the genuine edge: 200+ MW energised inside a year, a substation in under six months *(Crusoe dossier)*.

Its energy stack is the deepest published by any developer and effectively doubles as this report's supplier index: 29 GE Vernova LM2500XPRESS aeroderivative turbines at roughly 1 GW with five-minute starts *(Crusoe, 2025-01 — Crusoe dossier)*; ≈750 MW of Bergen reciprocating engines with Piller dynamic stabilisation for GPU transients *(Crusoe, 2026-06-03 — Crusoe dossier)*; **5 GW** of ON.energy medium-voltage battery "AI UPS" with ERCOT ride-through validated *(Crusoe, 2026-07-21 — Crusoe dossier)*; 12 GWh of Form Energy iron-air 100-hour storage from 2027 *(Crusoe, 2026-03-24 — Crusoe dossier)*; and nuclear optionality via Aalo, targeting a 10 MWe pilot at Idaho National Laboratory in 2027 and 50 MWe plants at Crusoe sites by end-2029 *(Crusoe, 2026-07-30 — Crusoe dossier)*.

The counterweight is concentration, and it has bitten twice. ANALYSIS – HIGH Customer concentration is the defining fragility, proven twice in 2026: Oracle and OpenAI halted the Abilene expansion, and Google managed Crusoe off the 1.8 GW Wyoming project over cost and timeline concerns — in both cases a single counterparty decision removed or reassigned gigawatt-scale work *(Crusoe dossier; Bloomberg via Yahoo, 2026-06-10 — Crusoe dossier)*. Contracted capacity is ≈4.9 GW against a claimed 40+ GW pipeline *(Crusoe, 2026-07 — Crusoe dossier)*, and a round was in talks at ≈$30B against roughly $500M of estimated 2025 revenue *(Bloomberg, 2026-07-02; Sacra estimates — Crusoe dossier)*. ANALYSIS – MODERATE The gas-heavy stack also carries accumulating regulatory exposure, with the Abilene plant running on minor permit-by-rule instruments and a pending major permit that would rank among the largest fossil plants in Texas *(Texas Tribune, 2026-07-09 — Crusoe dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Crusoe should sell time-to-power as a contracted product with a price and a penalty, rather than as a capability. Its whole edge is schedule, and schedule is currently being given away inside development agreements where a single customer can cancel and take the gigawatt with them. A guaranteed-energisation-date product, priced at a premium and backed by liquidated damages, is exactly what a buyer facing a four-to-eight-year interconnection queue will pay for — and it changes the negotiation from "we build for you" to "we sell you a date," which is far harder to walk away from. Second, diversify the counterparty base faster than the pipeline. Two customer decisions reshaped the company inside four months; the Childress structure with Lancium shows the model replicating with different partners *(Crusoe, 2026-07-15 — Crusoe dossier)*, and that replication is worth more than pipeline growth. Third, resolve the permitting position ahead of enforcement, not behind it. A merchant developer's franchise is its ability to keep siting plants; minor-permit pathways are a schedule loan whose interest is charged in future sites.

WHAT WOULD CHANGE THE VIEW

The 2026 revenue step. The valuation requires roughly a fourfold jump from ≈$500M as Stargate and Microsoft capacity converts; a shortfall means the marks are pricing a cloud company where the economics belong to a leveraged merchant developer that flips assets.

1.11 Equinix — the landlord that sets the colocation standard

WHAT THEY ARE

The largest carrier-neutral colocation operator, now rebuilding itself into an AI-capacity landlord — and the company whose specifications the rest of the colocation tier copies.

THE ENVIRONMENT

The strategic bet has been vindicated on the numbers but not de-risked. The June 2025 "Build Bolder" plan to roughly double capacity by 2029 cut long-term cash-flow-per-share growth guidance and knocked the shares about 18% in two days *(Motley Fool, 2025-06-26 — Equinix dossier)*. Thirteen months later Q2 2026 beat on both lines — revenue $2.625B (+16%) and adjusted funds from operations of $11.78 a share — with what management called the largest guidance raise in company history and capex stepping to **$5–7B a year** *(Q2 2026 results, 2026-07-29 — Equinix dossier)*. Capacity comes through vehicles that keep it off the balance sheet: a >$15B US joint venture with CPP Investments and GIC targeting more than 1.5 GW *(xScale JV, 2024-10-01 — Equinix dossier)*, and the ≈$4B atNorth acquisition bringing ≈800 MW installed in the Nordics with 1 GW of secured power *(atNorth acquisition, 2026-02-27 — Equinix dossier)*.

The second force is the constraint the whole sector shares, stated most clearly here. ANALYSIS – MODERATE Power scarcity, not demand or capital, is now the growth gate: 0.3% vacancy in Northern Virginia, Amsterdam's >70 MW connection moratorium, Singapore's rationed releases, and a record ≈$130B of blocked or delayed US projects in

Q1 2026 *(Equinix dossier)*. That is why the Nordic acquisition exists — it buys liquid-cooling-native capacity in a market where power is available.

Third, Equinix has crossed from power consumer to power developer, and its choices function as standards. It runs the colocation industry's largest fuel-cell fleet — Bloom units expanding past 100 MW across 19+ sites — alongside ≈1.25 GW of advanced-nuclear commitments and letters of intent including Oklo at 500 MW, the first colocation-operator small-modular-reactor deal *(Alternative-energy portfolio, 2025-08-14 — Equinix dossier)*. And the August 2026 Central Georgia EMC agreement is the most politically important structure in this chapter: a twenty-year take-or-pay in which *Equinix funds 100% of grid upgrades and overruns* plus up to $20M a year in property tax *(Central Georgia EMC agreement, 2026-08-06 — Equinix dossier)*. A take-or-pay obliges Equinix to pay for the power whether it uses it or not, which is precisely what makes a utility willing to build for it — and internalising the upgrade cost pre-empts the ratepayer backlash reshaping large-load tariffs nationally. ANALYSIS – HIGH Its choices are templates competitors and regulators now reference *(Equinix dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Equinix should standardise the megawatt and sell power as a product. Its two structural advantages are interconnection density — 513,000 interconnections is a moat pure capacity players cannot buy *(Equinix dossier)* — and the fact that it has already solved the political problem of large-load interconnection with the Central Georgia template. The commercial move is to package that: a repeatable AI-ready hall specification with direct-to-chip cooling, a defined on-site firm-power option, and a pre-negotiated ratepayer-protection structure, offered as one standard product across metros. Standardisation is what turns $5–7B a year of capex into predictable returns rather than 52 bespoke projects. Second, it should treat the take-or-pay structure as intellectual property and replicate it aggressively in every constrained jurisdiction before rivals do; being the operator regulators point to is worth more than any individual campus. Third, given that capex at this level against a roughly 1.8% dividend yield leaves no room for a demand pause, it should keep pushing capacity into joint ventures rather than onto its own balance sheet, even at the cost of some economics.

WHAT WOULD CHANGE THE VIEW

Bookings composition. Record gross bookings of $1.6B annualised with roughly 60% of the largest Q4 2025 deals AI-driven is what justifies the pivot *(FY2025 results, 2026-02-11 — Equinix dossier)*; if the AI share of large deals falls while capex stays at $5–7B, Equinix has traded real-estate-investment-trust economics for a growth market that stopped growing.

1.12 Constellation Energy — the electron seller who sets the price

WHAT THEY ARE

The largest US private-sector power producer and the owner of the largest US nuclear fleet — the company whose contracts define what firm, round-the-clock clean power costs a data center.

THE ENVIRONMENT

Constellation sells the scarcest product in this report and prices it accordingly. ANALYSIS – HIGH It has established the market-price ladder for firm clean power and that pricing power is its core asset: an analyst-estimated ≈$112/MWh for the Microsoft/Crane restart, ≈$70/MWh for Meta's existing-reactor output at Clinton, a management-stated $20–50/MWh premium band, and 920 MW of new fifteen-to-twenty-year contracts signed in a single quarter *(Constellation dossier; Energy Connects — Jefferies estimate, 2024-09; Utility Dive, 2025-06 — Constellation dossier)*. The mechanism behind the premium is simple and worth stating plainly: a data center that wants carbon-free power available every hour of every year cannot get it from wind and solar without storage, and there is only one existing technology that delivers it at gigawatt scale today. Scarcity of that attribute, not the cost of the fuel, sets the price.

The fleet got larger and less pure. The $16.4B Calpine acquisition created a ≈55 GW producer spanning ≈22 GW of nuclear running at 94.7% uptime plus gas and the largest US geothermal fleet *(Calpine completion, 2026-01-07 — Constellation dossier)*, after regulators forced ≈5.3 GW of divestitures to LS Power *(LS Power divestiture, 2026-03-18 — Constellation dossier)*. Q2 2026 adjusted earnings of $2.55 a share beat and full-year guidance rose to $11.50–12.50, with roughly 30% of clean baseload output now under long-term contract and the Crane restart clearing a FERC waiver and an NRC fuel amendment *(Q2 2026 results, 2026-08-05 — Constellation dossier)*. Crane itself is instructive on schedule: 835 MW restarting toward 2027 behind a $1B federal loan and, critically, 760 MW of capacity interconnection rights transferred from retiring units — which skipped a multi-year queue *(DOE $1B Crane loan, 2025-11-18 — Constellation dossier)*.

Two forces constrain it. `ANALYSIS — MODERATE` Political-backlash risk to capacity revenue is material and rising: three consecutive PJM capacity auctions clearing at the administrative cap, a thirteen-governor affordability coalition, and legislated clawback of capacity upside a live scenario for exactly the revenue behind the raised guidance *(Canary Media, 2025-07 — Constellation dossier)*. And `ANALYSIS — MODERATE` the ≈$43.75/MWh nuclear production-tax-credit floor is both foundation and cliff — it underwrites the protected-downside pitch through 2032, but expires inside the term of every contract now being signed *(S&P Global, 2024-02 — Constellation dossier)*. The upside catalyst is regulatory: FERC ordered PJM to write transparent co-location and large-load rules in December 2025, on Constellation's own complaint *(Utility Dive, 2025-12-18 — Constellation dossier)*.

RECOMMENDED STRATEGY

`ANALYSIS` *Desk view.* Constellation should sell the *package*, not the megawatt-hour. Calpine gave it something nuclear never could: dispatchable gas next to interconnection capacity and land, and the CyrusOne agreements — 380 MW plus an exclusive second 380 MW phase adjacent to the Freestone Energy Center, on top of roughly 400 MW at Thad Hill — show what that looks like when power, grid connectivity and site infrastructure are sold as one product *(CyrusOne Freestone agreement, 2026-02-09 — Constellation dossier)*. That bundle is worth far more than energy alone, because what the buyer is short of is a site with a live connection, not electricity. It should be the default offering across the whole Calpine footprint. Second, it should extend contract tenors past the tax-credit cliff now, while scarcity is at its peak and buyers are unusually willing — the credit's expiry is a known repricing event and the counterparty that locks price through it wins. Third, it should get ahead of the affordability politics with structures like Equinix's: where a data center funds the grid upgrades it causes, the clawback argument loses its best evidence.

WHAT WOULD CHANGE THE VIEW

PJM's co-location rules as filed. Management expects deal flow to begin with a bang once they land; rules that permit compensated co-location reopen behind-the-meter nuclear economics, while restrictive rules keep everything front-of-meter and cap the premium Constellation can charge.

1.13 Quanta Services — the installer becoming a manufacturer

WHAT THEY ARE

The largest electrical contractor in the United States — the company that physically builds the grid connections, substations and electrical systems every campus in this chapter depends on.

THE ENVIRONMENT

Quanta's environment is defined by owning the scarcest input in the buildout. It is the #1 US electrical contractor at $13.5B of electrical revenue, roughly **3.7 times the #2** *(EC&M 2025 Top 50, 2025-06 — Quanta dossier)*, and Q2 2026 revenue reached $9.557B (+41%, organic +27.4%) on a record $53.4B backlog with full-year guidance raised to $39.3–39.7B *(Q2 2026 results, 2026-07-30 — Quanta dossier)*. The scarcity is human: 69,500 employees of whom more than 50,000 are craft, over 80% of work self-performed, 15,500 hires in a trailing year, a 2,300-acre lineman college and roughly $250M a year on training — behind the chief executive's rule that **"it takes about four years to make a craftsman"** *(2026 Investor Day, 2026-03-31 — Quanta dossier)*. That four-year cycle is why labor cannot respond to a price signal the way steel can, and why the Technology and Load Centers end market can be guided to grow 220–240% in 2026 without competitors simply hiring their way in *(Q2 2026 call, 2026-07-30 — Quanta dossier)*.

The second force is a deliberate move up the supply chain into the equipment bottleneck itself: a $500–700M programme to nearly double high-voltage transformer manufacturing capacity by 2028, and the Hyosung HICO joint venture making gas circuit breakers up to 800 kV in Canonsburg, Pennsylvania — aimed squarely at breaker lead times of roughly **125 weeks** that are throttling load-center schedules *(2026 Investor Day, 2026-03-31; Q2 2026 release, 2026-07-30 — Quanta dossier)*. ANALYSIS — MODERATE This is strategically significant beyond Quanta: if the largest buyer of grid equipment self-supplies from 2028, the scarcity rents currently flowing to the transformer makers erode at the margin and Quanta's bids gain a schedule weapon competitors cannot match — though contractor-run manufacturing is a different discipline and there is no production track record yet *(Quanta dossier)*.

Third, the reported backlog is genuinely ambiguous and should be read in both directions. ANALYSIS — HIGH It understates the pipeline in two specific ways — the "significant majority" of the ≈3 GW NiSource combined-cycle programme is excluded, and data-center work runs book-and-burn through master service agreements outside backlog entirely — while also including estimates, with the formal SEC measure at only $33.6B *(Quanta dossier; NiSource award, 2025-10-30 — Quanta dossier)*. And ANALYSIS — MODERATE growth quality is the bear case: 41% headline growth with heavy acquisition assist, a wide gap between GAAP and adjusted earnings driven by deal amortisation, goodwill and intangibles at roughly 39% of assets, and a trailing multiple near 89 times after an intra-year drawdown from $789 to $363 that shows how violently that multiple compresses *(Quanta dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Quanta should sell schedule, priced explicitly, and keep integrating vertically to make that price credible. Its own arithmetic is the argument: at ≈$13.5M of craft-led spend per megawatt with 80–90% balance-of-plant coverage, Quanta is not bidding a scope, it is bidding a date — and a developer facing a 125-week breaker lead time will pay a premium measured in months of revenue, not in percentage points of construction cost. The transformer and breaker programmes exist precisely to make "we can energise you a year earlier" a supportable claim rather than a sales line; the follow-through is to price it that way in contracts, with schedule-linked terms rather than unit rates. Second, it should keep converting field labor into factory throughput, because that is the only lever that escapes the four-year constraint, and it should be honest that this — not acquisitions — is where durable margin comes from. Third, on capital discipline: with the multiple near 89 times and goodwill approaching two fifths of assets, the marginal acquisition is now more expensive to the equity story than organic craft hiring, which is the asset nobody can buy.

WHAT WOULD CHANGE THE VIEW

Organic growth excluding acquisitions. Q2 2026's organic 27.4% inside 41% headline growth is the number that matters; if organic growth converges toward the industry rate while acquisitions carry the headline, Quanta is a roll-up priced as a monopoly on craft labor.

1.14 Rosendin — the installer productising power

WHAT THEY ARE

The #2 US electrical contractor, employee-owned, and close to a pure-play on the AI data-center electrical build — now co-developing the power products it used to only install.

THE ENVIRONMENT

Rosendin's exposure is the most concentrated in this chapter. ANALYSIS — HIGH It is near-pure-play exposure to the AI data-center electrical buildout: revenue roughly doubled in two years from $2.80B in 2023 to $3.69B in 2024 to ≈$5.6B in 2025, moving it to #2 among US electrical contractors, with the growth timing matching the hyperscaler capex cycle exactly *(Rosendin dossier; EC&M 2025 Top 50, 2025-06 — Rosendin dossier)*. It reports 12,000+ employees and — remarkably for construction — customers providing work forecasts **ten to twenty years out** *(Construction Dive, 2026-01-28 — Rosendin dossier)*. Company-reported capability: 15+ GW of power deployed across data-center projects, $7.5B of total data-center work value, and 4M+ electrician man-hours of annual capacity *(Rosendin data-center sector page — Rosendin dossier)*, with a reference speed of 20 MW of electrical infrastructure delivered in 26 weeks on Project Goliath for CyrusOne *(Rosendin data-center sector page — Rosendin dossier)*.

The second force is the same labor ceiling Quanta faces, met with the same answer. ANALYSIS — MODERATE Prefabrication is the scaling strategy: 7,000+ MW of annual fabrication capacity across 2M+ square feet through Modular Power Solutions converts the binding constraint — skilled electricians — into factory throughput, and a robotic solar trial extends the same logic to field labor at 350–400 modules per eight-hour shift with a two-person crew, roughly triple manual rates *(Rosendin dossier; robotic solar installers, 2025-04-07 — Rosendin dossier)*.

Third, and most interesting commercially, Rosendin is moving from installing power to selling it. ANALYSIS — MODERATE The Energy Group is being deliberately converged with the data-center franchise: BESSUPS with FlexGen — a utility-scale battery engineered to replace data-center uninterruptible power supplies and diesel generators, sited outside the building at medium voltage, combining Rosendin's patents with FlexGen's controls intellectual property *(Business Wire, 2025-06-03 — Rosendin dossier)* — plus an aluminium-air backup partnership with Phinergy, and stated exploration of small modular reactors and gas turbines, read as a play to sell customers power capacity rather than just electrical installation *(Rosendin dossier; Energy Group leadership, 2026-02-19 — Rosendin dossier)*. It has the delivery record behind it: ≈9.5–10+ GWh of batteries installed and 151 substations across 330+ projects, company-reported, with individual projects corroborated by owners *(Rosendin renewable-energy sector page — Rosendin dossier)*.

RECOMMENDED STRATEGY

ANALYSIS *Desk view.* Rosendin should finish the move from contractor to product company, because it is the only path that escapes the margin ceiling of installation. BESSUPS is the right idea and it is under-commercialised: replacing the uninterruptible power supply and the diesels with a medium-voltage battery outside the building removes equipment, floor space and a maintenance regime from the data hall, and Rosendin is uniquely positioned to sell it because it is already the party the customer trusts to install the alternative. The move is to publish deployed references and a warranted availability figure — a product without a reference site is a patent, not a business. Second, the ten-to-twenty-year customer forecasts are an underexploited asset. That visibility justifies building fabrication capacity ahead of demand and pre-committing to long-lead equipment, which is exactly the schedule advantage the market pays for; a contractor that can promise a date because it already owns the factory slot and the switchgear is competing on something price cannot answer. Third, the employee-ownership structure is a genuine retention advantage in a labor-short market and should be marketed as a delivery-risk argument to customers, not merely as culture.

WHAT WOULD CHANGE THE VIEW

A named BESSUPS deployment. Commercialisation was announced in June 2025 with no deployed-site count published; a first reference installation at a named hyperscale campus would confirm that a contractor can sell a power product, while continued silence suggests the customers prefer their contractor to stay a contractor.

1.15 The coverage universe at a glance

The table below compresses all fourteen entries. Roles use the lens from section 10.0; strategies are this desk's analysis, not company guidance. ANALYSIS

COMPANY	ROLE	KEY EXPOSURE	DESK STRATEGY IN ONE LINE
NVIDIA	Specifier	Four direct customers at 61% of revenue; vendor-financed demand; China zeroed in guidance	Keep giving the power architecture away; grow demand it has not financed
Microsoft	Payer	No on-site UPS or generators at Fairwater Atlanta — wholly grid-dependent; firmness is nuclear and not yet delivered	Buy contracted flexibility, not batteries; underwrite third-party generation with the $678B backlog
Google	Payer + builder	Free cash flow negative; capex programme dependent on capital-market patience; CFE goal losing ground	Productise the energy park and sell firmness to others; buy duration, not more four-hour lithium
Amazon	Payer via counterparties	Equipment decisions sit with AES and Primergy; no campus-level storage at all; single-region concentration proven	Set one campus-level medium-voltage storage standard; sell firmness from the 40+ GW portfolio
Meta	Payer via utilities	Free cash flow of $784M; long-lived gas and nuclear against 15-year contracts; Louisiana docket	Rebalance firming scope into utility-tolled storage and contracted curtailability
OpenAI	Demand-setter	≈30 GW committed against ≈0.3 GW operational; circular financing; ≈$3.7B quarterly burn reported	Report energised megawatts, not announcements; sell curtailability for queue priority
xAI	Self-builder	$6.4B 2025 segment operating loss; $399M litigation accrual; mobile turbines must exit by July 2027	Capitalise the build machine as a merchant business; buy permits properly; tender the storage scope
Oracle	Landlord	−$23.7B free cash flow; ≈half of $638B RPO tied to one counterparty; 14–16% GPU-rental margins	Standardise to two power architectures; sell power alongside compute; disclose concentration
CoreWeave	Tenant	$24.9B debt, interest ≈24% of FY2025 revenue; leases transmit developers' schedules into guidance	Buy power capability not assets; price the Platinum operational record; publish realised rental rates
Crusoe	Developer	Two customer decisions reshaped it in 2026; ≈4.9 GW contracted vs 40+ GW pipeline; permit-by-rule exposure	Sell a guaranteed energisation date with penalties; diversify counterparties over pipeline

COMPANY	ROLE	KEY EXPOSURE	DESK STRATEGY IN ONE LINE
Equinix	Landlord	$5–7B/yr capex against ≈1.8% yield — no room for a demand pause; power scarcity gating growth	Standardise the AI-ready megawatt; replicate the ratepayer-protection take-or-pay everywhere
Constellation	Electron seller	PJM capacity revenue exposed to political clawback; ≈$43.75/MWh PTC (production tax credit) floor expires inside contract terms	Sell power + interconnection + site as one package; extend tenors through the tax-credit cliff now
Quanta	Installer + maker	≈89x trailing multiple; goodwill ≈39% of assets; backlog neither floor nor firm	Price schedule explicitly and back it with owned transformer and breaker capacity
Rosendin	Installer + product	Near-pure-play on hyperscaler capex; private and unaudited; BESSUPS has no published deployment	Publish a warranted BESSUPS reference; build fabrication ahead of the 10–20-year forecasts

1.16 What this means for your capital

The roles, not the logos, determine the return profile. Across these fourteen, three distinct investment characters emerge and they behave differently under the same shock. *Specifiers* (NVIDIA) capture the architecture rent and have the highest returns on capital, but their revenue quality is compromised by the degree to which they finance their own customers. *Payers* (Microsoft, Google, Amazon, Meta) are spending 2026 capex in the hundreds of billions, and the only durable differentiator among them is how much contracted revenue stands behind it — Microsoft's $678B backlog against Meta's $784M of quarterly free cash flow is the widest quality spread in the cohort, and it is not a small distinction. *Landlords, tenants and developers* (Oracle, CoreWeave, Crusoe, Equinix) are all levered plays on the same conversion assumption, which makes them highly correlated to one another and poorly diversifying. *Installers and electron sellers* (Quanta, Rosendin, Constellation) sit closest to the physical bottleneck and are the least dependent on any single buyer.

The mechanism that decides winners is who owns the scarce thing. Not compute — compute is being manufactured faster than it can be plugged in. The scarce things are an interconnection with a live date, a transformer or breaker slot, and a trained craftsman. Quanta owning the craft labor and buying the transformer and breaker capacity, Constellation owning the electrons and the interconnection rights, and Equinix owning the ratepayer-safe interconnection template are all the same trade expressed three ways. Every buyer in this chapter is ultimately paying one of them.

What to monitor, and on what timetable. Near term (this quarter and next): CoreWeave's Q2 print, due 11 August, and Microsoft's Azure growth rate, the one number its equity trades on. Through 2027: Abilene's progress toward the projected 1.2 GW, the Crane restart, the Louisiana Public Service Commission's decision on Entergy phase two, PJM's co-location rules as filed, and xAI's mobile-turbine removal deadline of July 2027. Structurally into 2028: Quanta's transformer capacity landing, Samsung SDI-class domestic cell capacity arriving, and the first Kyber-generation racks pulling ≈600 kW each — the moment rack-level storage stops being optional content.

ONE POLICY MECHANISM THAT TOUCHES THE DEVELOPER CHANNEL

Because Google, Amazon and Meta buy storage through developers and utilities rather than directly, what those counterparties are permitted to install matters to all three. On **28 July 2026** the FCC added foreign-produced connected power inverters to its Covered List, effective immediately; Covered-List products cannot receive FCC equipment authorization, which bars importing, marketing or selling them in the US. Two points are commonly misunderstood. It is *not* a China rule — it is an origin test, where "foreign-produced" means failing the Buy American domestic-end-product test (US manufacture with domestic component cost above 65% through 2028, rising to 75% in 2029), so an American brand manufacturing offshore is caught. And it is a *two-prong* test requiring both a bi-directional DC-AC or AC-DC device (microinverters, string inverters, central inverters, hybrid/battery-based inverters) *and* connectivity enabling remote communication, control or monitoring; the stated rationale is remote firmware push by a foreign adversary. It is **prospective only** — models authorized before 28 July keep authorization and may still be imported and sold, so this is a new-authorization freeze, not a market withdrawal. A conditional-approval path exists through the Department of War or DHS with applications to 1 January 2028, requiring ownership disclosure, a full bill of materials and supply-chain map, and a time-bound US onshoring plan; no review timeline has been published.

ANALYSIS The practical read for this chapter's buyers, and it is analysis rather than a sourced ruling: storage power-conversion systems and hybrid or battery-based inverters sit inside the scope, so the developer-owned utility-scale battery projects these hyperscalers contract for are directly affected on any *new* equipment authorization; a battery block shipped without a power-conversion system falls outside because there is no inverter in the scope of supply; and a unidirectional AC-DC server power supply or power shelf with wired management falls outside on a plain reading. The consequence to plan for is not a supply cliff but a narrowing of the qualified vendor list for new equipment models over 2027–28, which raises the value of domestic-content positions in exactly the channel Amazon, Google and Meta buy through.

What would change this framing. The whole chapter rests on one assumption: that announced capacity converts, slowly but reliably, into energised capacity. The observable that would break it is the order books of the equipment suppliers in sections 3 and 5 — because a buyer pausing cancels an order before it cancels an announcement. If two or more of GE Vernova, Hitachi Energy, Vertiv or ABB report a sequential decline in data-center orders while the buyers in this chapter keep guiding capex higher, believe the order books. They are the honest number, and they turn first.

This chapter is analysis for understanding commercial position. It is framing, not a recommendation to buy or sell any security.

2. Power, Grid and Prime Movers

Electricity reaches an AI chip only after a chain of conversions. It arrives from the transmission network at hundreds of kilovolts, is stepped down at a substation transformer, distributed around the campus at **medium voltage** — roughly 1 to 35 kV — stepped down again to **low voltage** (400–480 V), passed through switchgear and an uninterruptible power supply, carried along busway, and finally converted inside the rack to the fraction of a volt a processor uses. Every stage costs efficiency, copper, floor space and, above all, time. The ten companies in this chapter own different stages of that chain, and the two forces reshaping the market — physical scarcity and the 800 VDC architecture — hit each stage in opposite directions.

Three lanes, plus one company that sits outside the market entirely.

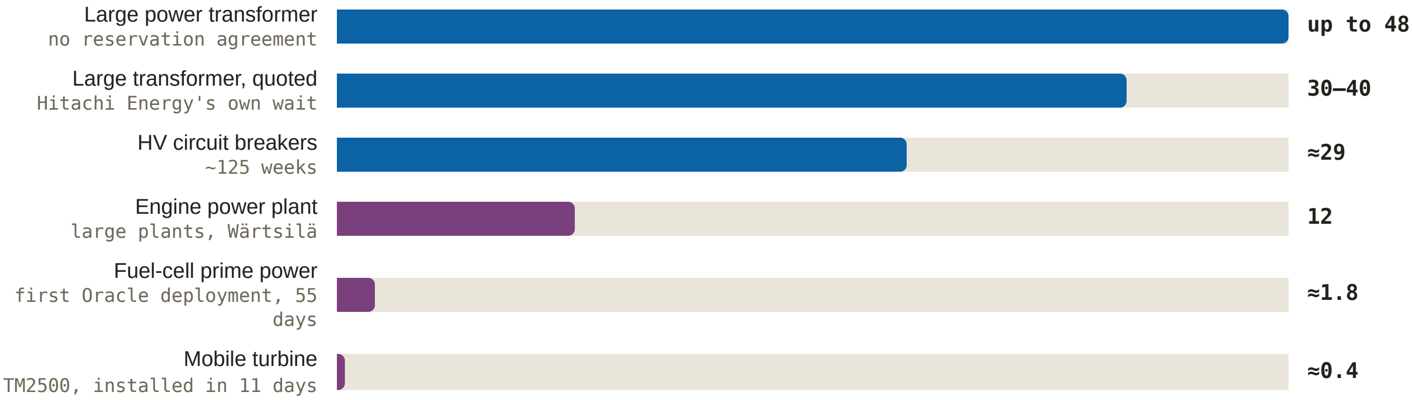
1. **The medium-voltage layer and the grid interface** — transformers, high-voltage switchgear, breakers, HVDC. Hitachi Energy, GE Vernova's Electrification segment, Siemens Energy's Grid Technologies, and ABB at the campus end. This is where the scarcity lives.
2. **Low voltage and the data hall** — centralized UPS, LV switchboards, busway, prefabricated power rooms, cooling. Schneider Electric, Vertiv, Eaton, and ABB's LV business. This is where 800 VDC does its damage.
3. **Prime movers** — the machines that turn fuel into electricity: gas turbines from GE Vernova and Siemens Energy, reciprocating engines from Wärtsilä, and, functionally, fuel cells from Bloom Energy. Sold on time-to-power.
4. **The excluded integrator** — Huawei Digital Power does all three lanes at once and cannot sell in the US.

WHY THE TWO FORCES POINT IN OPPOSITE DIRECTIONS

Scarcity rewards whoever owns a factory for a slow physical object. A large power transformer is not assembly-limited; it is limited by grain-oriented electrical steel, copper winding and a test bay. You cannot buy your way to one in a year. The same is true of turbine forgings. That makes the constraint a supply-side rent, and the rent accrues to the medium-voltage and prime-mover lanes.

800 VDC attacks the low-voltage lane. If a rack accepts 800 volts of direct current at its door, the LV AC switchboard and the centralized double-conversion UPS in their present form become unnecessary. Independent analysis projects the architecture *contracts* the centralized UPS market — a core Vertiv, Eaton and Schneider franchise — while creating roughly $11B of power-rack and $13B of solid-state-transformer demand *SemiAnalysis, 2025-10 — Vertiv dossier*. So for the grid houses 800 VDC is pure upside: they have no UPS franchise to lose. For the data-hall incumbents it is a **net-content question** (chapter 6) — do they capture more new content than they surrender? That question is the spine of every incumbent entry below.

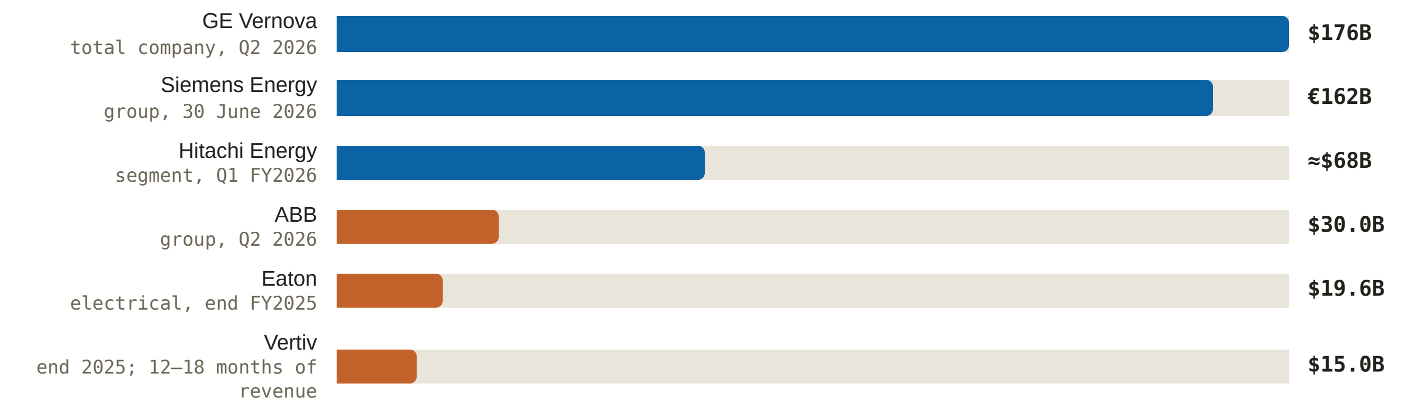
Figure 14: Time-to-power, by what you are buying (months, as reported)



Source: Nikkei Asia, 2024-05 — Hitachi Energy dossier; Quanta Q2 2026 release, 2026-07-30 — Quanta dossier; Data centre power solutions page — Wärtsilä dossier; Oracle 2.8 GW expansion, 2026-04-13 — Bloom Energy dossier; Gas power for data centers page — GE Vernova dossier. Weeks and days converted to months for comparability. The spread between the top bar and the bottom three is the arbitrage every company in this chapter is selling.

Read Figure 14 as the chapter's price list. A developer who has not reserved transformer capacity cannot energize a large campus before the end of the decade, which is why anything that compresses the schedule prices at a premium and why reservation agreements have become tradeable assets. Everything below is a variation on selling into, or around, that gap.

Figure 15: Forward work in hand — order backlog as reported ($/€ billion, not like-for-like)



Source: GE Vernova Q2 2026 press release, 2026-07-22; Siemens Energy Q3 FY26 earnings release, 2026-08-05; Investing.com, 2026-07-29 — Hitachi Energy dossier; ABB Q2 2026 press release, 2026-07-16; Eaton FY2025 results, 2026-02; Vertiv FY2025 results, 2026-02-11 — respective dossiers. Euros are not converted; scopes (total company, group, segment, divisional) and dates differ. Colour marks lane, not size.

The shape of Figure 15 is the shape of Figure 14. You carry an order book worth several years of revenue when your product takes three years to build — Hitachi Energy's backlog runs about 2.9x annual revenue *Hitachi Ltd FY2025 annual results presentation, 2026-04-27 — Hitachi Energy dossier*. A data-hall vendor's backlog covers twelve to eighteen months. Neither is better; they are different businesses, and they should not be valued with the same instinct.

2.1 GE Vernova

ENVIRONMENT

Spun out of General Electric in April 2024; roughly 85,000 employees and an installed base of about 7,000 gas turbines. FY2025 revenue $38.1B, orders $59.3B (+34% organic), free cash flow $3.7B *FY2025 press release, 2026-01-28 — GE Vernova dossier*. At Q2 2026: orders $24.2B (+88% organic), record backlog $176B, revenue $11.1B (+22%)

— and adjusted EPS of about $2.47 against roughly $3.04 expected, on tariffs of $100–200M and widening wind losses *Q2 2026 press release, 2026-07-22 — GE Vernova dossier*. It holds 116 GW of committed gas capacity: 53 GW of equipment backlog plus 63 GW of deposit-backed slot reservations now selling against **2031** delivery *Turbomachinery Magazine, 2026-07 — GE Vernova dossier*. Data-center Electrification orders ran ≈$0.7B in 2024, more than $2B in 2025, and more than $5B in the first half of 2026 alone *Q2 2026 press release, 2026-07-22 — GE Vernova dossier*. In February 2026 it paid $5.275B for the remaining half of Prolec GE — seven Americas transformer plants, five in the US, about 10,000 employees *Prolec GE acquisition completion, 2026-02-02 — GE Vernova dossier*.

A word on the mechanism that makes this company unusual. A **slot reservation** is a paid option on a manufacturing date: roughly 20% cash down, priced 10 to 20 points above the existing backlog, converting into a firm order later — Q2 2026 alone converted 10 GW *December 2025 Investor Update transcript, 2025-12-09 — GE Vernova dossier*. Buyers pay the premium because the alternative is an interconnection queue with no date at all. GE Vernova has turned a factory calendar into a financial product.

STRATEGY

Two franchises with very different durability, and — critically for the net-content question — **no centralized UPS business to defend**. GE Vernova attacks the data hall from outside it. It has published three AI-factory reference designs with NVIDIA that convert 13.8 kV AC to 800 VDC at the campus perimeter using solid-state transformers, and holds a development cost-share with a hyperscaler carrying a commitment to buy **1,000 SSTs from 2027 if the specification is met**; SSTs are described as 12–24 months from commercialization *AI Factory 800 VDC reference designs whitepaper, 2025-10-24; December 2025 Investor Update transcript, 2025-12-09 — GE Vernova dossier*. So the architecture that shrinks Vertiv's and Schneider's content is, for GE Vernova, an entirely new market it is bidding to create.

On turbines: `ANALYSIS` (High) the pricing power is real but it is scarcity rent, not a durable moat — reservations imply roughly $3,000/kW combined-cycle pricing, about triple three years ago, while output ramps from ≈20 GW/yr toward 24 GW in 2028 and actions toward 30 GW by 2030, and competing supply lands in the same 2028–2030 windows GE Vernova is now selling *GE Vernova dossier*. Management expects to be largely sold out of 2030 deliveries by the end of 2026 *Utility Dive, 2025-12 — GE Vernova dossier*. On transformers: `ANALYSIS` (High) the more defensible franchise, with industry lead times of 2.5–4 years modeled to persist toward 2030 and Prolec's US plants now wholly owned *GE Vernova dossier; Wood Mackenzie, 2024-07*. The aeroderivative line is the third leg, sold explicitly as schedule: LM6000 five-minute starts, LM2500XPRESS above 99.8% availability, TM2500 mobile units installed in 11 days, with Chevron's "power foundries" targeting up to 4 GW for data centers by 2027 *Gas power for data centers page; Chevron data-center power partnership, 2025-06-15 — GE Vernova dossier*.

WHAT WOULD CHANGE THIS VIEW

The reservation book is the tell. Watch conversion rate and cancellations: `ANALYSIS` (Moderate) backlog-to-revenue conversion is the central risk the headline numbers hide, since 116 GW sits upstream of interconnection queues and gas infrastructure, and 20% deposits mitigate but do not eliminate slippage *GE Vernova dossier*. A slipped SST specification would move the $13B solid-state-transformer opportunity to challengers instead. And a second consecutive quarter of record orders alongside missed earnings would make the gap between the order narrative and the earnings record the story rather than a footnote.

2.2 Siemens Energy

ENVIRONMENT

The other Western giant that gates both scarce inputs, and the only vendor that can book turbines, transformers, switchgear, HVDC and grid-stability equipment for a gigawatt campus from one order. Q3 FY2026 was the best quarter in its history: orders €17.9B, revenue €11,447M (+18.5% comparable), net income €1,188M against €983M expected, the highest quarterly group margin ever at 14.2%, and a record €162B backlog. Gas Services alone booked €10.0B of orders — a **book-to-bill of 2.65**, 73 turbines totalling 15 GW — with US data centers the named driver; Grid Technologies booked €5.4B (+28%) at a 19.9% margin on a €51B backlog, transformers the largest growth contributor *Q3 FY26 earnings release, 2026-08-05 — Siemens Energy dossier*. Unit sales nearly doubled in a year: 194 turbines in FY25 against 100 in FY24 *Siemens Energy release, 2025-11-14*. Large-frame slots are nearly full through 2028 and slot-reservation *fees* are back for the first time since the early-2000s gas boom *POWER Magazine, 2025-02 — Siemens Energy dossier*.

Book-to-bill of 2.65 means that for every euro of revenue recognised, €2.65 of new work arrived. That is a company filling its future faster than it can empty it — which sounds unambiguously good and is not. Revenue is capacity-limited, so the excess lengthens the queue rather than accelerating cash. Two pieces of history belong in the frame: Siemens Gamesa's 2023 blade and bearing defects forced a €2.2B charge and a roughly €4.5B net loss *CNBC, 2023-08-07 — Siemens Energy dossier*, and the €7.5B German federal counter-guarantee was fully replaced by a €9B private bank facility in June 2025, two years early *Energy Connects, 2025-06 — Siemens Energy dossier*. The company rebrands to **Omterra** from late 2026 — the Siemens name is licensed and time-limited *Siemens Energy release, 2026-07-14*.

STRATEGY

Deliberate restraint, and it is the single most consequential supply decision in this chapter. Capacity in transformers and large gas turbines expands 30–50% across FY26–28, alongside a €6B buyback and up to €10B of shareholder returns, with the FY28 margin target raised to 14–16% *CMD 2025, 2025-11-20 — Siemens Energy dossier*. ANALYSIS the choice of measured expansion plus returns supports pricing power but *prolongs the bottleneck* *Siemens Energy dossier*. Localization is the tariff-and-lead-time hedge: $1B in the US, a Mississippi HV switchgear plant, Charlotte gas-turbine manufacturing resuming, and a $150M Charlotte large-power-transformer plant against US Southeast lead times of 36-plus months, with roughly 80% of US large power transformers imported *Siemens Energy $1B US investment release, 2026-02-03*.

On net content, Siemens Energy is in the same privileged position as GE Vernova and solves the data-hall problem by partnership rather than product: ANALYSIS (High) the Eaton alliance's 500 MW SGT-800 block is the most productized onsite-power offering from any major OEM — redundant turbines plus battery storage, no diesel gensets, 99.99%-plus availability, and a claim of up to two years off time-to-power *Siemens Energy release, 2025-06-03; Siemens Energy dossier*. Siemens Energy supplies generation and the grid interface; Eaton carries the low-voltage exposure. ANALYSIS (Moderate) the grid side is the more durable franchise, since transformer scarcity outlasts any single generation cycle *Siemens Energy dossier*. Utility demand corroborates the turbine side — Xcel ordered ten SGT6-5000F units, 2,088 MW, for Texas *Siemens Energy release, 2025-10-01*.

WHAT WOULD CHANGE THIS VIEW

ANALYSIS (Moderate) valuation is the principal risk to reading momentum as strength: roughly 70x earnings, a sell-side target range from €84 to €175, and a stock that fell on a 21% earnings beat *Siemens Energy dossier; Capital.com valuation debate — Siemens Energy dossier*. Two observables would break the thesis. Gas Services book-to-bill falling toward 1.0 would say the demand shock has passed its peak. Conversely, capacity expansion accelerated beyond the stated 30–50% would signal the OEMs have begun a supply race — good for buyers, corrosive to the rent. A wind relapse would not threaten the company as it did in 2023, but it would dent the FY28 margin path ANALYSIS (Low) *Siemens Energy dossier*.

2.3 Hitachi Energy

ENVIRONMENT

The world's largest transformer maker by installed base and the leader in HVDC converter stations at about 15.3% share, having supplied roughly half of all HVDC projects ever built *Mordor Intelligence — Hitachi Energy dossier*. FY2025: revenue $19.8B (+26%), orders $32.8B — up from $12.4B in FY21 — backlog $57.9B, and an adjusted EBITA margin of 13.4% against 6.1% four years earlier *Hitachi Ltd FY2025 annual results presentation, 2026-04-27 — Hitachi Energy dossier*. Q1 FY2026 pushed further: segment revenue ¥911.9B (+37%), orders up 87%, backlog above ¥10 trillion, roughly $68B *Investing.com, 2026-07-29 — Hitachi Energy dossier*. Its own large-transformer waits run 30–40 months, up to four years without a reservation *Nikkei Asia, 2024-05 — Hitachi Energy dossier*. Wood Mackenzie models a roughly 30% US power-transformer deficit in 2025, prices up 77% since 2019, and constraints persisting well into the 2030s *Wood Mackenzie, 2025-02 — Hitachi Energy dossier*.

The E.ON framework — up to $700M for more than 20,000 transformers under a capacity-reservation model — is first-party proof that transformer capacity is now allocated years ahead by contract *Hitachi Energy release, 2025-07-28*. That mechanism deserves a moment: a reservation agreement converts a queue position into a contract, which turns a physical constraint into a balance-sheet asset for the seller and a real option for the buyer. It is the same instrument GE Vernova sells on turbines, applied to the tighter bottleneck. The supply answer is real and late: a $9B-plus global programme headlined by a $457M South Boston, Virginia plant billed as the largest US transformer factory, **operational 2028**, with Alamo, Tennessee at $106M and CAD 270M in Québec *Hitachi Energy release, 2025-09-04*.

STRATEGY

ANALYSIS (High) this is scarcity monetization executed cleanly — orders 2.6x and margin more than doubled in four years, backlog at about 2.9x revenue, and no major project-loss charges in the press record *Hitachi Energy dossier*. On net content the position is the purest in the chapter: Hitachi Energy has *nothing* in the data hall to lose and is reaching for the interface, co-designing grid-to-rack conversion for 800 VDC racks from 3 MW to gigawatt class, with a company-published and independently unverified claim of a fifteen-fold power-supply increase *Hitachi Energy release, 2025-10-13*. ANALYSIS (Moderate) its edge is owning the grid side of that interface; its gap is having no UPS or rack-power installed base to sell through *Hitachi Energy dossier*. The near-term commercial product is speed: Grid-eXpand prefabricated modular grid-connection substations, deployed for Kauri CAB's urban Frankfurt AI campus at 110 kV — Europe's tightest data-center power market *Hitachi Energy release, 2026-07-13*.

The company also authored the sector's volatility thesis. Its CEO's warning that AI training loads spike "up to 10 times" normal draw *The Logic, 2025-01 — Hitachi Energy dossier* was corroborated by August 2026 Bloomberg reporting that volatile AI demand is physically damaging data-center equipment *Bloomberg, 2026-08-06 — Hitachi Energy dossier*. The mechanism matters commercially: transformers and switchgear are designed around a duty cycle, and repeated fast swings age insulation and mechanical parts. That is the demand driver beneath every stabilization and buffering product in this report.

WHAT WOULD CHANGE THIS VIEW

Lead times shortening materially before the 2028 capacity wave would deflate the rent early. ANALYSIS (Moderate) cyclical risk is real but back-loaded — roughly $1.8B of announced US transformer investment across the industry all lands 2027–28, Wood Mackenzie sees the deficit shrinking by 2030, and a contrarian reading already exists that the problem is procurement inefficiency rather than a shortage *Hitachi Energy dossier; POWER Magazine, 2026-01*. One structural caveat for an investor: this is not directly purchasable. It is a Hitachi subsidiary, and the only listed pure proxy, Hitachi Energy India, trades at 116x FY27 earnings after a 107.7% year with two-thirds of its backlog HVDC-dependent ANALYSIS (Low) *MarketScreener, 2026-07 — Hitachi Energy dossier*. The franchise is not separately listed: it sits inside Hitachi Ltd, and the only listed pure proxy for it is Hitachi Energy India.

2.4 ABB

ENVIRONMENT

The hottest data-center order book of any diversified electrical. Q2 2026: record group orders of $12.0B (+28% comparable), Electrification orders of $7.2B (+60%) — the first quarter ever above $7B — triple-digit data-center order growth, Americas Electrification up 114%, an Electrification margin of 24.9% and a segment backlog of $13.7B (+57%); group backlog $30.0B, operational EBITA $1,925M at 20.2% *ABB Q2 2026 press release, 2026-07-16*. FY2025 closed at record orders of $36.8B (+17%) and an all-time-high 19.0% operational EBITA margin, with roughly $600M of individual data-center orders above $100M in Q4 alone *ABB release, 2026-02-09*. Roughly 40% of ABB's electrification research now targets next-generation data-center technology. Position honestly stated: top five in data-center power, behind Schneider overall and behind Schneider, Vertiv, Eaton and Huawei in UPS *MarketsandMarkets — ABB dossier*.

Two products explain why it matters more than that ranking suggests. **HiPerGuard** is the industry's first static medium-voltage UPS — 2.5 MW per unit, hard-parallel to 25 MW, voltage classes from 6 kV to 24 kV plus a new 34.5 kV class that connects directly to the grid without a step-down, 98% efficiency, intrusive maintenance only every ten years *HiPerGuard product and technical data pages; ABB Q2 2026 press release, 2026-07-16 — ABB dossier*. The mechanism is the whole point: a conventional UPS sits at low voltage, so every protected megawatt must first be stepped down, rectified, inverted and redistributed. Putting the UPS at medium voltage deletes a transformation stage from a 100 MW campus. It is deployed at Applied Digital's Polaris Forge 2 in 25 MW blocks against 300 MW *ABB release, 2025-11-25*. And **SACE Infinitus** is the world's first IEC-certified solid-state circuit breaker *ABB release, 2025-10-13* — necessary rather than clever, because direct current has no zero-crossing at which to extinguish an arc, so DC faults must be cleared in microseconds by semiconductors rather than milliseconds by moving contacts. Solid-state breakers are a prerequisite for 800 VDC, not an accessory.

STRATEGY

On the net-content question ABB has the most comfortable answer of the four data-hall incumbents, precisely because it has the least to defend. Its LV UPS franchise — MegaFlex DPA, expanded to 1,000–2,000 kW per system *ABB release, 2026-03-23* — is the smallest of the four, while its replacement content is the most differentiated. ANALYSIS (High) ABB is winning the medium-voltage layer: HiPerGuard has no direct static MV UPS competitor at scale, the 34.5 kV class removes a conversion stage entirely, and the NVIDIA 800 VDC role makes it the architecture incumbent for the next rack generation, with Eaton, Schneider and Vertiv competing at low voltage *ABB dossier*. It has also turned load volatility into hardware: VoltaGrid has ordered 62 flywheel synchronous condensers across two tranches, the first operational in April 2026 *ABB release, 2025-11; Engineering.com, 2026-03-25*. A synchronous condenser is a spinning mass that supplies inertia and short-circuit strength without net energy — a shock absorber for a weak island grid. ANALYSIS (High) those orders are the strongest independent validation that AI load volatility is now a purchasable product category, though undisclosed unit values leave revenue materiality unproven *ABB dossier*.

Two standing facts belong in any assessment. ANALYSIS (High) ABB sold its grid and transmission business to Hitachi before the supercycle, so it cannot supply the transformers or HVDC its own campuses' grid connections require; and it is a three-time US bribery settler, most recently the 2022 $315M Kusile/Eskom resolution *ABB dossier; DOJ, 2022-12-02*.

WHAT WOULD CHANGE THIS VIEW

ANALYSIS (Moderate) capital allocation is now the biggest swing factor: Robotics was sold to SoftBank for $5.375B and the proceeds recycled into a $5.5B offer for Rotork that the market called pricy — shares fell about 4% on the announcement — with roughly $14B of stated remaining firepower against a P/E near 37 and a third-party consensus rating the dossier records as neutral *ABB dossier; ABB release, 2025-10-08; ABB Q2 2026 press release, 2026-07-16*. Watch for a credible competing static MV UPS at scale, or for hyperscalers standardising on low-voltage feeds into 800 VDC racks without an MV UPS stage — either would dissolve the differentiation. One more expensive acquisition would do the rest.

2.5 Schneider Electric

ENVIRONMENT

Shares the global number-one position in data-center physical infrastructure with Vertiv, the two separated by about 0.1 point *Dell'Oro, 2026-06 — Schneider dossier*. H1 2026 was the strongest half in its history: revenue €21.23B (+14.0% organic), a record Q2 of €11.46B (+16.5%), North America up 23%, adjusted EBITA €4.09B (+22.1% organic) at a 19.3% margin, 120 basis points better, against roughly €3.8B expected — with data-center and semiconductor demand both growing triple-digit and full-year guidance raised to +14–19% organic EBITA growth *H1 2026 results release, 2026-07-30 — Schneider dossier*. FY2025 revenue was €40.15B with record free cash flow of €4.64B, though both divisions missed divisional EBITA expectations and the group print was rescued by central costs *FY2025 results release, 2026-02-26 — Schneider dossier*. Capacity: more than $700M across eight US sites through 2027 *US $700M investment plan, 2025-03-25*. Liquid cooling came via Motivair, 75% for $850M *Motivair acquisition announcement, 2024-10-07*.

Its distinctive asset is not a box. It is the only incumbent with NVIDIA co-developed designs spanning electrical, cooling controls and whole-factory blueprints: GB200 NVL72 at 132 kW racks, GB300 NVL72 at 142 kW racks and 1,152 GPUs per hall with the industry-first integrated power-and-cooling controls design interoperable with NVIDIA Mission Control, and gigawatt AI-factory blueprints with ETAP electrical models and AVEVA embedded in NVIDIA's Omniverse DSX Blueprint, including Vera Rubin NVL72 *GB300 NVL72 + Mission Control reference designs, 2025-09-18; Gigawatt AI-factory blueprints (GTC 2026), 2026-03-16 — Schneider dossier*. The mechanism is worth internalising because it is the most transferable idea in this chapter: if the single-line diagram a customer starts from already contains your switchgear and your UPS, the bid you eventually win was decided before it was issued. That is a distribution advantage disguised as engineering.

STRATEGY

The net-content question lands hardest here, and the design franchise is what buys time to answer it. Schneider has the largest centralized-UPS franchise at risk of anyone in the tier — the Galaxy VXL runs 500–1,250 kW per frame from 125 kW modules, parallelable to 5 MW at up to 99% efficiency in eConversion mode *Galaxy VXL launch,* *2024-12-02* — and its 800 VDC answer, a sidecar converting AC to 800 VDC for racks up to 1.2 MW with integrated energy storage for backup and load smoothing, carries **no commercial date** *800 VDC sidecar announcement, 2025-10-13*. So the honest reading is: Schneider is currently defending, not capturing, and its defence is the specification layer. ANALYSIS (High) the edge is the design layer rather than the hardware layer — a position that pulls its hardware into projects before competitive bidding starts — with the caveat that no design-linked order figures are published and conversion is inferred from triple-digit demand growth and a $373M Digital Realty supply-capacity agreement *Schneider dossier*. ANALYSIS (High) the two-speed portfolio is both stabilizer and discount: triple-digit data-center demand against Industrial Automation margins that missed at 14.2% in FY2025 cushions an AI-capex shock better than a pure play but caps the multiple *Schneider dossier*. The software layer — AVEVA, ETAP, the pending Cognite acquisition — has no direct equivalent at Vertiv or Eaton *Schneider dossier*.

WHAT WOULD CHANGE THIS VIEW

One observable dominates: a commercial date for the sidecar. If a rival ships a named 800 VDC switchboard into a hyperscaler design while Schneider's remains undated past Kyber in 2027, the design franchise will have failed to convert into the new content, and the shrinking UPS market becomes the whole story. Watch also for Schneider's role in NVIDIA reference designs narrowing, and for data-center growth decelerating while industrial automation stays soft — the two-speed cushion only works if one speed is fast. ANALYSIS (Moderate) the Motivair-led cooling lead is contestable, with Vertiv assembling equivalent scope in smaller pieces and Eaton paying $9.5B for Boyd *Schneider dossier*.

2.6 Eaton

ENVIRONMENT

The most aggressive M&A conversion in the sector: roughly $12.5B committed in fourteen months across Fibrebond at $1.45B, Ultra PCS at $1.55B, and Boyd Thermal at $9.5B — 22.5x forward EBITDA — earnings before interest, tax, depreciation *and* amortisation, a wider measure than the EBITA quoted for Hitachi Energy in chapter 4, so the two margins are not directly comparable *Fibrebond acquisition close, 2025-04-01; Boyd Thermal close, 2026-03-12; MarketScreener, 2025-11 — Eaton dossier*. FY2025 revenue $27.4B (+10%), adjusted EPS $12.07, electrical backlog up 29% to $19.6B, with data-center orders up 200% in Q4 *FY2025 results, 2026-02 — Eaton dossier*; data-center orders then grew 240% year over year in Q1 2026. Q2 2026 revenue was $8.53B (+21.4%) against $8.01B expected — but a third of that growth was acquired, +14% organic and +7% M&A — with adjusted EPS of $3.15, guidance raised to $13.40–13.60, and Electrical Global backlog up 103% *Q2 2026 results release, 2026-07-31; Zacks/Yahoo Q2 2026 verdict, 2026-07 — Eaton dossier*. The Mobility group separates by Q1 2027, completing the conversion to an electrical and aerospace pure play.

Two products teach mechanisms worth knowing. Fibrebond and the Flexnode collaboration deliver turnkey prefabricated data halls of 3.5–35 MW supporting rack densities above 1 MW, with roughly 35% average schedule compression against stick-build *Flexnode collaboration, 2026-01-28 — Eaton dossier* — moving competition from component price to *installed-megawatt lead time*. And EnergyAware, a grid-interactive UPS co-developed with Microsoft and deployed at Bahnhof in Sweden, lets a UPS fleet sell frequency regulation while still protecting the load *Eaton dossier*. That is the mechanism by which the UPS and battery-storage lanes converge: a battery idle 99.9% of the time is a stranded asset, and letting it bid into ancillary-services markets — the grid operator's markets for frequency regulation and reserve, which pay for the *capability to respond* rather than for energy delivered — turns insurance into revenue.

STRATEGY

Eaton states the net-content problem more plainly than any peer. ANALYSIS (Moderate) the 800 VDC transition is net-neutral-to-positive *only if its replacement content wins*: the architecture shrinks the centralized UPS franchise, and Eaton's hedge — XLM supercapacitor fast-cycle backup with roughly twenty-year life, ORv3 busbar distribution, and Resilient Power solid-state conversion — is announced but not yet a shipping 800 VDC switchboard, a gap independent analysis notes for all AC-switchboard incumbents *Eaton dossier; 800 VDC reference architecture (OCP), 2025-10-13*. The packaging answer is Beam Rubin DSX, end-to-end power and cooling aligned to NVIDIA's Vera Rubin DSX design with Omniverse digital-twin assets *Eaton dossier, 2026-03-16*. ANALYSIS (High) the broader conversion — components incumbent to delivered-power-systems vendor — is inorganic by design, selling scope breadth rather than component price *Eaton dossier*.

One discipline lesson matters more than any product. ANALYSIS (High) the data-center order surge reflects content-per-megawatt expansion and lumpy multi-year hyperscaler frame agreements more than unit-share capture — the underlying market grows about 7.5% annually while Eaton's data-center orders grew 200–240%, third-party share structures are unchanged, and rolling-twelve-month organic order growth of roughly 38–42% is the cleaner metric *Eaton dossier*. Any triple-digit order figure in this sector should be read the same way. Eaton also holds the generation-agnostic seat: the Siemens Energy alliance means it sells the electrical scope of a behind-the-fence campus whichever prime mover wins *Siemens Energy alliance, 2025-06-03 — Eaton dossier*. And it productized the volatility problem first, adding market-first detection of subsynchronous oscillations caused by AI GPU load bursting, pushed to the installed base by remote firmware update *Eaton release, 2025-09-09* — a rhythmic exchange of energy below grid frequency that can build rather than damp when load steps in bursts.

WHAT WOULD CHANGE THIS VIEW

ANALYSIS (Moderate) execution risk is concentrated in 2026–2027, with four integrations and the Mobility spin landing together against a stock priced for perfection, a widening GAAP-to-adjusted gap from acquisition amortization, and a Q3 2026 GAAP guide below consensus *Eaton dossier*. Falsifiers: Boyd accretion slipping past the promised year two; organic order growth decelerating toward the market rate while acquired revenue carries the print; and a rival shipping a named 800 VDC switchboard with Eaton still pre-product. ANALYSIS (Moderate) backlog duration is two-sided — only about 20% of the 307 GW US pipeline converts near-term, so the order book gives visibility but concentrates sensitivity to any capex pause, and would show it first *Eaton dossier*.

2.7 Vertiv

ENVIRONMENT

The pure play, and therefore the tier's most sensitive instrument: roughly three-quarters or more of revenue is data-center. FY2025 revenue $10.23B (+27.7%), adjusted EPS $4.20 (+47%), organic orders up about 252% in Q4 at a book-to-bill near 2.9x, and backlog of $15.0B — up 109% and covering twelve to eighteen months of forward revenue *FY2025 results, 2026-02-11 — Vertiv dossier*. Then Q2 2026 cracked the streak: revenue of $3.27B (+24%) missed roughly $3.37B — the first miss after six beats — on supply-chain congestion and multi-phase project timing, even as adjusted EPS of $1.52 beat, operating margin reached 22.6% (+410 basis points), free cash flow rose 234% and full-year guidance was raised to $13.8–14.2B. The shares fell about 10–17% in a session *Q2 2026 results, 2026-07-29; StockStory Q2 2026 verdict, 2026-07-29 — Vertiv dossier*. Dell'Oro places it in a statistical tie with Schneider for number one in data-center physical infrastructure *Dell'Oro, 2025-06 — Vertiv dossier*.

ANALYSIS (High) Vertiv functions as the sector's leveraged proxy, amplifying AI-capex sentiment in both directions: up 14–20% on the Q4 2025 order print, down 25–30% within days on the DeepSeek scare, down 10–17% on a 3%

revenue miss — against a business whose backlog and book-to-bill kept strengthening throughout *Vertiv dossier*. The mechanism is concentration. When three-quarters of revenue depends on one capital cycle, the equity carries that cycle's second derivative, not its level. That is a useful property to understand and a dangerous one to mistake for a view on the underlying market.

STRATEGY

Vertiv has the most centralized-UPS content at risk in the sector and the most aggressive replacement roadmap — the net-content question in its most acute form. The Trinergy franchise runs 1,500–2,500 kVA blocks scalable to 20 MW at 97% or better efficiency, explicitly pitched at fluctuating AI loads *Trinergy UPS product page — Vertiv dossier*; that is precisely the content 800 VDC removes. Against it, the full 800 VDC ecosystem — centralized rectifiers, DC busway, rack DC-DC converters and DC-compatible backup — releases in **2H 2026**, ahead of NVIDIA's Kyber generation in 2027 *800 VDC engineering readiness with NVIDIA, 2025-10-13 — Vertiv dossier*, and PowerDirect Rack, an OCP ORv3-compliant 50V DC shelf covering 33–132 kW per rack, already puts Vertiv in the shelf and battery-backup territory where the ODM tier operates *PowerDirect Rack / SmartRun / Unify launch, 2025-03-18 — Vertiv dossier*. ANALYSIS (Moderate) the transition remains a net-content question Vertiv has not yet proven, since no incumbent has named a shipping 800 VDC switchboard *Vertiv dossier; SemiAnalysis, 2025-10*.

It is also crossing lanes. EnergyCore Grid is utility-grade storage scaling from 1 MW to more than 200 MW with integrated power conversion and energy management, black-start capable *EnergyCore Grid BESS launch, 2025-12-03 — Vertiv dossier*. ANALYSIS (Moderate) that signals convergence between data-center power vendors and the storage integrator space, attacking the interconnection-queue problem from the data-center relationship rather than the independent-power-producer channel *Vertiv dossier* — which is the real lesson: in campus storage, channel decides more than product. Prefabrication is the third leg, with OneCore delivering 5–50 MW factory-assembled blocks and SmartRun installing at roughly 1 MW per day, up to 85% faster than stick-build *OneCore prefab launch, 2025-08-05; PowerDirect Rack / SmartRun / Unify launch, 2025-03-18 — Vertiv dossier*.

WHAT WOULD CHANGE THIS VIEW

ANALYSIS (Moderate) execution, not demand, is the binding risk into 2027 — five plant expansions and multiple acquisition integrations running at once, against the 2021–22 precedent when management underestimated inflation, produced a 37% one-day drop and drew securities litigation *Vertiv dossier; Rosen Law — Vertiv dossier*. Two clean observables. If the 800 VDC portfolio ships on schedule in 2H 2026 and is named inside a hyperscaler design, the net-content question resolves in Vertiv's favour and the largest overhang on the pure play lifts. A second consecutive revenue miss makes the Q2 print structural rather than timing, and the leveraged-proxy property will work in the unhelpful direction.

2.8 Bloom Energy

ENVIRONMENT

A solid-oxide fuel-cell maker that became an AI-infrastructure supplier in about two years. Q2 2026 was its first billion-dollar quarter: revenue $1.065B (+165.5%) against $814M expected, non-GAAP EPS $0.78, GAAP operating income $182.2M, cash of $2.67B, with full-year guidance raised twice to $3.9–4.2B — roughly 100% growth at midpoint *Q2 2026 results, 2026-07-28 — Bloom Energy dossier*. The Oracle master agreement reaches up to 2.8 GW, with up to 2.45 GW earmarked for Project Jupiter at Doña Ana, New Mexico, *displacing previously planned gas turbines and diesel gensets*, and the first Oracle deployment delivered in 55 days against a 90-day commitment *Oracle 2.8 GW expansion, 2026-04-13; Oracle Project Jupiter fuel-cell announcement, 2026-04-27 — Bloom Energy dossier*.

AEP firmed 1 GW at roughly $2.65B *AEP $2.65B firming, 2026-01-15 — Bloom Energy dossier*, Brookfield's financing framework expanded fivefold to $25B *Brookfield $25B expansion, 2026-06-30*, and the AI segment now runs about 250 MW across roughly 24 customers from near zero two years ago *MiTAC expansion, 2026-08-06 — Bloom Energy dossier*. Fremont capacity doubles from 1 GW to 2 GW by the end of 2026 *Bloom Energy dossier*.

What a solid-oxide fuel cell is matters commercially. It oxidizes fuel electrochemically rather than burning it, so there is no flame and almost no nitrogen oxides or particulates — 0.003 lbs/MWh of NOx, with 65% falling to 53% LHV electrical efficiency over life, about 100 MW per acre, and no water consumption in normal operation *Energy Server 6.5 datasheet, 2026-02 — Bloom Energy dossier*. The schedule advantage often comes from that chemistry rather than from the efficiency: a non-combustion device clears air permitting that a turbine fleet or a diesel farm does not. The tradeoff is stated in the same datasheet — 679 to 833 lbs/MWh of CO_2 on natural gas, broadly comparable to a modern combined-cycle plant.

STRATEGY

ANALYSIS (High) the core sale is time, not electricity: 55–90 day deployments against interconnection waits of up to roughly seven years, with AEP framing fuel cells as bridge prime power at six to nine months versus about three years for turbines while transmission and 2030s nuclear catch up *Bloom Energy dossier; Utility Dive, 2024-11*. The AEP order also supplies the sector's best public pricing benchmark for fuel-cell prime power at about $2.65 per watt ANALYSIS *Bloom Energy dossier*. A quieter architectural point: a fuel cell produces direct current natively, and every shipment is now 800V DC-ready *Energy Server 6.5 datasheet, 2026-02 — Bloom Energy dossier*. In an 800 VDC campus that removes an inversion stage outright — a genuine efficiency and capital saving, not marketing.

The discipline required here is separating framework from firm. ANALYSIS (High) the headline pipeline overstates firm demand: Oracle's 2.8 GW is a master-agreement ceiling with roughly 1.2 GW contracted, AEP's 1 GW was only 100 MW firm in its first year, the ~$20B backlog splits roughly $6B product and $14B service, and management itself disclaims more than about six months of demand visibility *Bloom Energy dossier; Bloom Q4/FY2025 results, 2026-02-05*. ANALYSIS (Moderate) manufacturing ramp is the binding constraint and the scandium question is unresolved — a July 2026 short report argued that 5 GW of output would require roughly 220 tonnes a year of scandium oxide, near projected global supply, against management's rebuttal of 25 GW supported from industrial tailings; neither side has published verifiable sourcing data, a securities class action was filed on 31 July 2026, and the shares fell about 38% in the following month amid roughly $44M of ninety-day insider selling *Bloom Energy dossier; 24/7 Wall St, 2026-07-09; Bloom Energy dossier, 2026-07-31*. The ageing bear case is the counterweight: ANALYSIS (Moderate) service margins moved from −20.6% in FY2023 to +10% GAAP in FY2025, answering the historical degradation critique directly *Bloom Energy dossier*.

WHAT WOULD CHANGE THIS VIEW

Three observables. The Oracle framework converting materially below the ~1.2 GW already contracted would confirm that the ceiling was the story. Litigation discovery producing bill-of-materials disclosure that establishes a hard scandium ceiling would cap the ramp regardless of demand. And a competing prime mover matching sub-90-day *permitted* deployment would remove the moat, because the moat is the permit, not the machine. ANALYSIS (Low) natural-gas anchoring is a medium-term reputational and regulatory exposure rather than a demand constraint — no customer loss or rule change has yet been tied to it *Bloom Energy dossier*.

2.9 Wärtsilä

ENVIRONMENT

A Finnish marine and energy group founded in 1834, about 17,900 employees, now running two divergent stories. The retained business builds power plants from medium-speed four-stroke engines: 6–23 MW generating sets scaling past 450 MW per plant, with the 31SG at 10.8–12.4 MW and up to 52.1% electrical efficiency, the 46TS family at 20.4–22.9 MW with two-minute starts, and the 50SG at 18.4 MW with **30-second grid connection**; large plants are operational in twelve months, use under five litres of water per hour, and are claimed to burn 20–35% less fuel than gas turbines *Engine power plant products page; Data centre power solutions page — Wärtsilä dossier*. US data-center demand is converting: 789 MW of first US data-center projects in 2025, followed by a 429 MW order for 24 × 50SG engines with commercial operation late 2028 to early 2029 *Financial Statements Bulletin 2025, 2026-01; 429 MW US data-centre power plant order, 2026-01-29 — Wärtsilä dossier*. Group H1 2026: order intake €4,934M (+23%, an all-time high), net sales €3,004M (−1%), comparable operating result €411M at 13.7%, order book €8,976M (+13%) *Half-year Financial Report Jan–Jun 2026, 2026-07-21 — Wärtsilä dossier*.

The engine-versus-turbine tradeoff is the reason this lane exists beside two turbine giants. A gas turbine is one large rotating machine: highest output per unit and best efficiency at full load and large scale, but efficiency falls away at part load and starting takes minutes. A plant assembled from twenty 20 MW engines is granular — you run only the units you need, so part-load efficiency holds; one unit failing costs 5% of output rather than 50%; and a 30-second grid connection means the plant can follow a load that steps. For an AI campus whose draw swings by design, granularity is worth more than peak efficiency.

STRATEGY

This is the chapter's clearest example of a company choosing where *not* to compete. ANALYSIS (High) the storage joint venture is a managed retreat from merchant integration: 2025 order intake collapsed 60% to €455M and 52% to 2,677 MWh on US tariffs, FEOC rules and price competition, making standalone scale uneconomic *Wärtsilä dossier; Financial Statements Bulletin 2025, 2026-01*. The entire Energy Storage business — Quantum hardware, the GEMS control platform, lifecycle services, a €719M order book and roughly 480 staff — folds into a 50/50 joint venture with Germany's RCT Solutions, announced 15 June 2026 and reclassified as discontinued operations, with closing expected in Q3 2026 *Energy Storage JV with RCT Solutions, 2026-06-15 — Wärtsilä dossier*. The cost is disclosed and the payback is deferred: a guided €40–50M hit to the 2026 operating result, with the JV loss-making in 2026 and positive results expected in late 2027; analysts cited weak profitability, "very limited" engine-storage synergies and no disclosed proceeds *Half-year Financial Report Jan–Jun 2026, 2026-07-21; Energy-Storage.News, 2026-06 — Wärtsilä dossier*.

Read as capital allocation, the logic is coherent: keep the lane where the engineering is scarce and the service annuities are long, exit the lane where cells and price set the outcome and where policy decides who may bid at all. ANALYSIS (High) data-center exposure continues regardless of the storage outcome, through the retained engine business *Wärtsilä dossier*. Two assets survive the change of ownership. ANALYSIS (Moderate) the cybersecurity-first positioning is durable differentiation — a first-in-industry IEC 62443 certification for GEMS, with GEMS Cloud Connect achieving SOC 2 Type 1 in February 2026 and Type 2 underway — and an increasingly required procurement checkbox for utilities and data-center buyers *Wärtsilä dossier; GEMS Cloud Connect SOC 2 Type 1, 2026-02-19*. Wood Mackenzie's inaugural integrator ranking still placed Wärtsilä in the global top ten as a pure-play *ESS News, 2026-07-14 — Wärtsilä dossier*. The fuel path is the long option: the 31SG-H2 runs natural gas with up to 25% hydrogen by volume before conversion, and the pure-hydrogen 31H2 launched in June 2024 with deliveries from 2026 *World's first 100% hydrogen-ready engine power plant, 2024-06-18 — Wärtsilä dossier*.

WHAT WOULD CHANGE THIS VIEW

US data-center engine bookings failing to compound beyond the 789 MW and 429 MW base would say the lane is a niche rather than a franchise — the 2028–29 commercial-operation dates mean the next two years of orders, not revenue, are the evidence. On the other side, the JV missing its promised late-2027 profitability would recast the exit as mistimed rather than well-timed. ANALYSIS (Moderate) counterparty transition around the Q3 2026 close is the live operational risk, and contract and service continuity for the delivered fleet is the thing to watch *Wärtsilä dossier* — which is also why rivals treat it as a displacement window.

2.10 Huawei Digital Power

ENVIRONMENT

The largest China-based competitor across the whole AIDC power stack, and the one company here an American investor cannot own and an American buyer cannot choose. 2025 segment revenue was CNY 77.3B, roughly $11.2B, up 12.7% — about 9% of Huawei group revenue and its second-strongest growth engine after automotive, though growth halved from 24.4% in 2024, and no product-line breakdown is disclosed *Huawei 2025 Annual Report, 2026-03 — Huawei Digital Power dossier*. Positions, stated fairly: number one in prefabricated modular data centers per Omdia, though absent from the North American and Western European regional podiums *Omdia, 2024-07 — Huawei Digital Power dossier*; number one in PV inverters at 176 GWac shipped in 2024 *Wood Mackenzie, 2025-06 — Huawei Digital Power dossier*; top five in data-center UPS behind Schneider, Vertiv and Eaton *MarketsandMarkets — Huawei Digital Power dossier*; and joint fourth in utility BESS integration in 2024, third in Europe in 2025 *Wood Mackenzie, 2026-04 — Huawei Digital Power dossier*.

Its flagship products are the pattern the industry is converging on. PowerPOD is a 40-foot containerized power block at 2.4 MW standard or 3.2 MW customized, more than 90% prefabricated, cutting delivery from 28 to 18 weeks with 70% less onsite labor *Global Data Center Facility Summit 2025, 2025-05-13 — Huawei Digital Power dossier*. SmartLi 3.0 is an LFP battery backup unit at 6C discharge with pack-level fire extinguishing and a 70% smaller footprint than lead-acid *SmartLi lithium BBU product page — Huawei Digital Power dossier*. The LUTERRA grid-forming storage platform, launched June 2026, offers plant-level grid forming with inertia across a 0–20 second time constant at 5 ms response and minute-level black start of gigawatt-hour plants *LUTERRA grid-forming ESS launch, 2026-06-23 — Huawei Digital Power dossier*. And the Red Sea project remains the off-grid reference: 400 MW of PV plus 1.3 GWh of storage, 100% renewable, 21-plus months at 99.9% critical-load reliability *ESS News, 2024-09-18 — Huawei Digital Power dossier*.

THE FCC COVERED LIST ACTION — THE CORRECTED POSITION, AND WHAT IT ACTUALLY CATCHES

On **28 July 2026** the FCC added foreign-produced connected power inverters to its Covered List, effective immediately. Covered-List products cannot receive FCC equipment authorization, which bars importing, marketing or selling them in the US *FCC Fact Sheet, 2026-07-28 — Huawei Digital Power dossier*. *Policy in this report is current to 16 August 2026 and was re-verified for this edition; the previous edition described this as a pending, draft, China-specific rule, which was wrong.*

It is not a China rule. It is an origin test. "Foreign-produced" means failing the Buy American domestic-end-product standard — US manufacture with domestic component cost above 65% through 2028, rising to 75% in 2029. An American brand manufacturing offshore is caught; Taiwanese and Korean production is caught on identical terms *FCC Covered List action, re-verified 16 August 2026 — not a dossier source*.

It is a two-prong test and both prongs must be met: (a) a *bi-directional* device converting DC to AC or AC to DC — enumerated as microinverters, string inverters, central inverters, and hybrid or battery-based inverters; and (b) connectivity enabling remote communication, control or monitoring via Wi-Fi, cellular, Bluetooth "or another similar connection." The stated rationale is remote firmware push by a foreign adversary.

It is prospective only. Models authorized before 28 July keep their authorization and may still be imported and sold. This is a new-authorization freeze, not a market withdrawal. A conditional-approval path exists through the Department of War or DHS, with applications open to 1 January 2028, requiring ownership disclosure, a full bill of materials and supply-chain map, and a time-bound US onshoring plan; no review timeline has been published *Cooley, 2026-07-31; Morgan Lewis, 2026-08 — Huawei Digital Power dossier*. Equipment lacking remote control, or using an air-gapped external control architecture, is reported to fall outside.

What that means mechanically, on a plain reading. A unidirectional AC-DC server power supply or power shelf managed over wires — PMBus, or IPMI, Redfish or SNMP over Ethernet — falls outside, because it fails both prongs. A DC battery block shipped without a power-conversion system falls outside too, because there is no inverter in the scope of supply. Hybrid and battery-based inverters, storage power-conversion systems and EV-charging equipment are inside. Whether wired management satisfies "another similar connection" is unresolved and determines how far the rule reaches into wired-managed power electronics *Energy-Storage.News, 2026-07 — Huawei Digital Power dossier*. Any read of a specific company's exposure is analysis, not a sourced ruling.

STRATEGY

ANALYSIS (High) Huawei Digital Power's competitive fate in AIDC is set by regulation, not engineering: the US is closed by the Entity List since 2019, NDAA 889, and now the Covered List inverter prohibition; EU-funded energy projects phase Chinese inverters out with full restrictions on new contracts by **April 2027**; and the security rationale travels with every grid-connected product. Its addressable AIDC market is structurally the Middle East, APAC, Africa and Latin America *Huawei Digital Power dossier; Euronews, 2026-05-04*. The evidentiary backbone is documented, not rhetorical: May 2025 reporting found undisclosed communication hardware in Chinese-made inverters and batteries, capable of bypassing utility firewalls, with a November 2024 remote-disable incident cited as precedent *Utility Dive, 2025-05-14 — Huawei Digital Power dossier*.

ANALYSIS (High) for Western-market purposes Huawei matters as the *excluded benchmark*: PowerPOD's prefabrication, SmartLi's density and the 18-week delivery claim set the specification and speed bar that Vertiv, Eaton, Schneider and Delta are measured against in international bids, and the exclusion is a durable share subsidy for those vendors in NATO geographies *Huawei Digital Power dossier*. Be equally fair about the limits of the marketing. ANALYSIS (Moderate) the gap between claim and position is widest in storage: joint fourth globally in 2024, third in Europe in 2025, and it lost its Middle East stronghold — BYD and Sungrow took 87% of 2025 regional share, and Sungrow, not Huawei, won the 7.8 GWh Saudi SEC project next door to Huawei's own Red Sea showcase. The grid-forming engineering is genuinely first-rate; the commercial position is mid-pack *Huawei Digital Power dossier; ESS News, 2025-12-18*. ANALYSIS (Moderate) with growth decelerating and no product-line split disclosed, every sub-segment momentum claim should be treated as unverifiable from company channels — the opacity itself is the finding *Huawei Digital Power dossier*.

What to take from a strong competitor you cannot buy. Three things, and none of them is a position. First, it is a reference price and a reference schedule, not a holding: when a Western vendor's delivery or density claim looks impressive, the correct comparison is the excluded benchmark, not the vendor next door. Second, exclusion is a revenue subsidy *with a duration*, and durations should be priced. The Entity List and NDAA 889 are open-ended; the Covered List action is prospective with a conditional-approval window open to 1 January 2028; the EU phase-out is dated April 2027 and applies to funded projects, not private ones. The subsidy is therefore durable in the US and dated in Europe — and a margin derived from policy should not carry the same multiple as a margin derived from engineering. Third, the customer's inability to choose has a cost. Where Huawei is available, a buyer gets one integrated stack on an 18-week clock; in the US the same buyer assembles a multi-vendor solution. Some of the schedule pressure the US market feels is the price of exclusion — which is not an argument against the policy, but it does identify the West's integration gap as a real, quantified competitive cost that whoever closes it first can monetize. ANALYSIS (Moderate) the Covered List action reshapes the Western AIDC supply chain more than it changes Huawei's own position: Huawei was already excluded, so its incremental loss is minimal, while Sungrow — which retains US market access — absorbed the shock, shedding roughly CNY 100B (about $14.8B) of market value over the following month *Huawei Digital Power dossier*.

WHAT WOULD CHANGE THIS VIEW

Watch the conditional-approval docket. A granted approval with a credible onshoring plan — for anyone, not necessarily Huawei — would reopen a lane the market currently assumes is shut, and would compress the policy premium the Western vendors earn. A published FCC clarification that wired management does satisfy "another similar connection" would widen the rule sharply into rack power and storage conversion, with consequences well beyond this company. In the other direction, an EU decision extending the April 2027 restrictions from funded projects to private ones would shrink the addressable map again. And if Chinese domestic AIDC demand accelerates while segment growth keeps decelerating, the constraint will have shifted from policy to execution.

2.11 The tier at a glance

COMPANY	LANE	PRINCIPAL EXPOSURE	STRATEGY IN ONE LINE
GE Vernova	Prime movers + MV grid (Americas transformers)	Turbine slot rents and transformer capacity; no UPS franchise to defend; earnings quality	Sell the factory calendar as a reservable product, own the American transformer toll booth, and attack the data hall from outside via solid-state transformers
Siemens Energy	Prime movers + MV grid, one order book	The same two rents, plus a wind residual and a ~70x multiple	Expand 30–50% rather than maximally, return capital, and productize onsite power with a partner who carries the data-hall risk
Hitachi Energy	MV grid and HVDC — the interface	The transformer bottleneck itself; no rack or UPS installed base; not directly listed	Ration capacity by reservation, monetize the calendar, and reach to the rack through the NVIDIA architecture

COMPANY	LANE	PRINCIPAL EXPOSURE	STRATEGY IN ONE LINE
ABB	MV inside the campus, plus LV	Smallest LV UPS franchise of the four incumbents; capital allocation; no transformer or HVDC scope	Win medium voltage where no static competitor exists, sell grid stability as hardware, and buy growth carefully
Schneider Electric	LV and the data hall, plus the design layer	Largest centralized-UPS franchise at risk; an undated 800 VDC product; two-speed portfolio	Own the reference design so the bid is decided before it is issued
Eaton	LV and the data hall, plus prefab and on-site generation scope	Centralized UPS at risk; four integrations and a spin-off landing at once	Buy the scope, sell installed megawatts rather than components, and hedge the DC transition with supercapacitors and busbar
Vertiv	LV and the data hall — pure play	Most centralized-UPS content at risk; execution at scale; concentration	Ship the full 800 VDC portfolio first, and cross into utility-scale storage through the data-center relationship
Bloom Energy	Prime movers, non-combustion	Firm-versus-framework demand; manufacturing ramp; unresolved scandium sourcing	Sell weeks instead of years, and let utilities and Brookfield carry the balance sheet
Wärtsilä	Prime movers (engines); exiting storage	US data-center engine bookings; joint-venture transition risk	Keep the lane where the engineering is scarce, exit the one where price and policy decide the winner
Huawei Digital Power	All three lanes — outside the West	Policy, not competition; storage position weaker than the marketing; disclosure opacity	Compound in the Middle East, APAC, Africa and Latin America, and set the specification bar everyone else is measured against

What this means for your capital

This tier separates into **rent collectors** and **re-architects**, and the two require different questions.

The rent is real and it is visible in margins, not just in narrative: Hitachi Energy's adjusted EBITA margin went from 6.1% to 13.4% in four years *Hitachi Ltd FY2025 annual results presentation, 2026-04-27 — Hitachi Energy dossier*, and GE Vernova sells reservations at 10–20 points above its own backlog *December 2025 Investor Update transcript, 2025-12-09 — GE Vernova dossier*. But the rent has a published expiry. Hitachi's South Boston plant is operational in 2028; Siemens Energy's 30–50% capacity expansion runs across FY26–28; GE Vernova reaches 24 GW of output in 2028 and takes actions toward 30 GW by 2030; and roughly $1.8B of announced US transformer investment across the industry all lands 2027–28 ANALYSIS (Moderate) *Hitachi Energy dossier*. So the question is not whether the pricing power exists — it does, and it is documented — but what multiple is appropriate for a cash flow whose supply relief already has a date attached. Rents earned against a dated constraint should not be capitalized like franchises.

For the re-architects the question is narrower and resolves sooner. Vertiv's full 800 VDC portfolio is due in 2H 2026; NVIDIA's Kyber generation arrives in 2027; GE Vernova's 1,000-unit solid-state-transformer decision falls in 2027; Schneider's sidecar has no date at all. The single most informative event in this chapter over the next four quarters is the **first named, shipping 800 VDC switchboard**. If it comes from an incumbent, the roughly $11B power-rack and $13B solid-state-transformer opportunities largely accrue to the companies that already own the customer relationship, and the shrinking centralized-UPS franchise is a manageable substitution. If it comes from the grid tier or the ODM challengers, the incumbents will have surrendered content without capturing the replacement, and the design-layer advantages that Schneider and Eaton lean on will have bought delay rather than position.

Three mechanisms are worth monitoring on a calendar. **Time-to-power is the pricing variable** — every product in this chapter is priced off the spread in Figure 14, so any credible compression of transformer or breaker lead times before 2028 deflates the whole tier's pricing assumptions simultaneously. **Volatility has become a product category**: Hitachi's CEO diagnosed it, Bloomberg documented the equipment damage, ABB sold 62 synchronous condensers, and Eaton shipped oscillation detection by firmware update. When a physical problem turns into a purchasable line item, a new content stream appears — and that transition, not the diagnosis, is the investable moment. **Order books are the early-warning instrument, and must be read organically.** Eaton's own analysis is the template: data-center orders up 200–240% against an underlying market growing about 7.5% a year means content-per-megawatt and frame agreements, not share capture, and rolling-twelve-month organic growth is the honest number ANALYSIS (High) *Eaton dossier*. With only about 20% of the 307 GW US pipeline converting near-term ANALYSIS (Moderate) *Eaton dossier*, the order book is where a capex pause would appear first — well before revenue.

What would change the whole chapter's view. Any one of five observables: a shipping 800 VDC switchboard named inside a hyperscaler design; transformer lead times shortening materially before the 2028 capacity wave; Gas Services or reservation book-to-bill falling toward 1.0 at either turbine house; a second consecutive revenue miss at Vertiv, converting a timing blip into evidence of execution strain; or a hyperscaler capex guide cut, which would test the assumption that backlog equals demand rather than intention.

This passage is framing for understanding a sector's mechanics. It is not investment advice and contains no recommendation to buy or sell any security.

3. Storage, Cells and Rack Power

These sixteen companies all sell something that stores or converts electricity for a data center, and they are routinely lumped together as "battery companies." That grouping destroys the information you need. Sort them by layer first, by where they physically manufacture second, and by nationality last — or not at all. Policy in 2026 tests products and places, not passports.

THE FOUR LAYERS

1. Cell makers. A cell is the smallest sealed electrochemical unit — one can of chemistry with two terminals. Cells are sold by the ampere-hour (Ah), which measures charge, not energy; energy in watt-hours is Ah times voltage, so a 314 Ah lithium-iron-phosphate (LFP) cell at 3.2 V holds about 1 kWh. This is process manufacturing: huge fixed cost, thin unit margin, returns driven by yield and factory utilisation. It is also the layer where material provenance is documented — which is why nearly every content rule and sanction lands here.
2. DC-block makers. A block is a factory-assembled enclosure of cells, busbars, liquid cooling, fire suppression and a battery management system, trucked to site and bolted to a pad. It is "DC" because it contains no inverter — it stores and releases direct current only. Bigger blocks mean fewer deliveries, fewer field connections and less scarce electrician time per MWh, which is why the industry sprinted from 3.35 MWh to 14.5 MWh per unit in three years.
3. AC-system integrators. Grids run on alternating current. Someone must add the power conversion system (PCS — the bidirectional inverter that turns DC into grid-synchronous AC and back), the medium-voltage transformer and switchgear, the plant controller, and then warrant the whole assembly for twenty years. The warranty is the product. Integrators are paid to absorb performance risk, and they are who a developer or hyperscaler actually signs.
4. Rack-power ODMs. Inside the data hall the power path is different again: utility AC steps down, then converts to the low DC voltage a GPU consumes. The firms building that chain — power shelves, power supply units (PSUs), battery backup units (BBUs), supercapacitor shelves, DC-DC bricks, solid-state transformers (SSTs) — are contract designers and manufacturers to NVIDIA's rack specification. Their competitive unit is watts per cubic inch and qualification date, not dollars per kWh.

612 GWh	14.5 MWh	≈76%	480 kW
Cell layer — the 2025 ESS cell market, up 94.6% as glut flipped to tightness	Block layer — largest single factory-built DC block (BYD HaoHan), vs a 6–7 MWh norm	AC-system layer — Chinese integrators' share of a market above 100 GW/yr	Rack layer — BBU capacity embedded in one Delta 660 kW in-row power rack

Source: InfoLink via ESS News, 2026-02-09 — Hithium and EVE dossiers; pv magazine, 2025-09-19 — BYD dossier; Wood Mackenzie via ESS News, 2026-07-14 — Sungrow dossier; Delta 800 VDC deep-dive, 2026-05-28 — Delta dossier.

WHY LAYER AND FOOTPRINT BEAT NATIONALITY

Four separate policy instruments act on this tier, and each one tests a different thing. Foreign-entity-of-concern and domestic-content rules test the ownership and material lineage of a component, which puts the burden on the cell layer. Tariffs test the point of import, so a Chinese firm assembling blocks in Texas from Chinese cells still pays the cell duty while gaining a "US-assembled" claim. The FCC's Covered List action tests what a product functionally is, plus where it was made. Chinese export controls test whether technology crosses a border. Mechanics for all four are in the policy chapter (see the policy chapter); what matters here is that they can point in opposite directions for the same company.

THE OPERATIVE TEST, IN ONE PARAGRAPH

On 28 July 2026 the FCC added foreign-produced connected power inverters to its Covered List, effective immediately; Covered-List products cannot receive equipment authorization, which bars importing, marketing or selling them in the US. It is not a China rule — "foreign-produced" is an origin test, failing the Buy American domestic-end-product standard (US manufacture with domestic component cost above 65% through 2028, rising to 75% in 2029), so an American brand manufacturing offshore is caught and Delta and LITEON are foreign-produced too. Both prongs must be met: a bidirectional device converting DC-AC or AC-DC (microinverters, string inverters, central inverters, hybrid and battery-based inverters), and connectivity enabling remote communication, control or monitoring. It is prospective only — models authorized before 28 July keep authorization and may still be sold. A conditional-approval path runs through the Department of War or DHS, applications to 1 January 2028, with no published review timeline. (see the policy chapter)

Read that against the layer map and the consequences fall out. A unidirectional AC-DC server PSU or power shelf with wired management (PMBus, IPMI/Redfish/SNMP over Ethernet) falls outside on a plain reading. A DC block shipped without a PCS also falls outside, because no inverter is in the scope of supply. Hybrid and battery-based inverters, storage PCS and EV-charging equipment are inside. So the same company can hold two products with opposite exposure — Sinexcel is the cleanest illustration, and Delta the largest.

THE MIRROR-IMAGE CONSTRAINT — AND WHY WHOLLY-OWNED PLANTS ARE THE ANSWER

US policy pushes Chinese suppliers toward local production and compliant structures. Chinese policy pushes back on the cheapest way to get there. MOFCOM — China's Ministry of Commerce — added LFP and LMFP (lithium manganese iron phosphate, an energy-denser LFP variant) cathode preparation technology to its restricted-technology catalogue on 15 July 2025, and MOFCOM/Customs Announcement No. 58 of 9 October 2025 (effective 8 November 2025) added dual-use controls on lithium batteries and on cathode and graphite anode materials (Chinese export-control dates re-verified 16 August 2026 — not dossier sources). The effect is not a ban but a discretion: licensing your process to a third party abroad becomes a permission you must be granted, not a deal you can sign. Hithium's cell-technology licence to India's Reliance Industries collapsed in January 2026 under exactly that tightening, and Reliance halted its India cell plan *Indian Chemical News, 2026-01-12 — Hithium dossier*; a second Indian licensing arrangement, Amara Raja's with Gotion, ended the same way (re-verified 16 August 2026 — not a dossier source). A wholly-owned overseas plant transfers nothing to anyone — the technology stays inside the group. That is why the localisation you see from Chinese leaders is own-name factories and assembly lines rather than licence-and-royalty structures. ANALYSIS (High)

3.1 Sungrow — the integrator that took both crowns

Sungrow sits at the AC-system layer and is now ranked first in both of Wood Mackenzie's 2026 comprehensive rankings — BESS integrator and PV inverter — the first vendor ever to hold both and the first to take the integrator spot from Tesla *ESS News, 2026-07-14 — Sungrow dossier*. Storage became the profit centre in FY2025: ESS revenue of RMB 37.29B (+49.4%) was 41.8% of the group at a 36.5% gross margin, overtaking inverters, on 43 GWh of shipments (+53.5%) *EnergyTrend, 2026-04-01 — Sungrow dossier*. PowerTitan 3.0 is built on the first mass-producible 684 Ah cell with a silicon-carbide PCS, reaching 12.5 MWh per 30-ft unit *pv magazine, 2025-06-10 — Sungrow dossier*. Anchor references: 7.5 GWh plus 2.6 GW of inverters for Masdar's 5.2 GW Abu Dhabi project *ESS News, 2026-05-22 — Sungrow dossier* and the 7.8 GWh Algihaz plant in Saudi Arabia, described as the world's largest grid-forming storage plant *pv magazine, 2024-07-16 — Sungrow dossier*. Q1 2026 was ugly: revenue RMB 15.56B (−18.3%), net profit RMB 2.29B (−40.1%) *Smartkarma, 2026-04 — Sungrow dossier*.

STRATEGY

Sungrow is the most acutely policy-exposed name in the file, because its product is the inverter — the exact object the Covered List captures. Its storage PCS and hybrid inverters are bidirectional with remote monitoring, meeting both prongs, and its manufacturing is Chinese, so it is foreign-produced under the origin test; the EU is separately phasing Sungrow and Huawei inverters out of funded projects by April 2027 *Euronews, 2026-05-04 — Huawei dossier*. ANALYSIS (Moderate) The bind has no licensing escape, because there is nothing to licence: Sungrow buys cells and specifies its own format rather than making them, so MOFCOM's cathode controls barely touch it while the inverter rules hit it hardest. The available paths are wholly-owned non-China manufacturing it has not announced, and pre-28-July authorizations that carry existing models. Its own stated proof points are European: roughly 10 GWh of PowerTitan 3.0 deliveries in 2026 and the re-filed A+H listing that would fund the build.

3.2 Tesla — standardisation and software instead of density

Tesla is the leading Western AC-system integrator, ranked second behind Sungrow in the 2026 comprehensive ranking *ESS News, 2026-07-14 — Tesla dossier*, having deployed a record 46.7 GWh in 2025 (+49%) with a 28.7% Q4 segment margin *Teslarati, 2026-01 — Tesla dossier*. Megapack 3 delivers about 5 MWh on a Tesla-designed 2.8-litre LFP cell with a silicon-carbide inverter and 78% fewer thermal connection points, shipping as one 39-tonne piece with no onsite assembly; production started at Brookshire on 6 August 2026, sixteen months from groundbreaking, against 50 GWh/yr of designed capacity *Electrek, 2026-08-06 — Tesla dossier*. Megablock packages four of them with an integrated medium-voltage transformer and switchgear: 20 MWh, 91% round-trip efficiency at medium voltage, 23% faster installation, up to 40% lower construction cost, 1 GWh deployable in about 20 business days *ESS News, 2025-09-09 — Tesla dossier*. First Megapacks with US-made LFP cells shipped in July 2026 from the roughly 7 GWh Sparks plant, built with CATL equipment *TeslaNorth, 2026-07-31 — Tesla dossier*.

STRATEGY

Tesla's answer to 12.5 and 14.5 MWh Chinese containers is not density but standardisation plus software — one mass-manufacturable unit, site-level integration in Megablock, and the Autobidder/Opticaster trading layer ANALYSIS (Moderate). Its compliance position is bought rather than owned: the $4.3B LFP cell award to LG Energy Solution for Lansing from 2027 *Electrek, 2026-03-17 — LGES dossier* and cells multi-sourced across the US, Southeast Asia and China. ANALYSIS (Moderate) Onshored LFP reduces but does not eliminate structural dependence on CATL, and the Sparks line was built with CATL equipment — a durable regulatory advantage requires that the rules keep testing where cells are made rather than whose process makes them. Treat the reported roughly 25 GWh NatPower framework as unconfirmed European volume ANALYSIS (Low).

3.3 CATL — the leader whose localisation route is now discretionary

CATL is the number one energy-storage battery maker for five consecutive years, holding 27.1% of global ESS batteries on 125 GWh in H1 2026 *CnEVPost, 2026-08-04 — CATL dossier*, with the ESS line at RMB 53.26B in H1 2026 (+87.5%, 24.0% gross margin) and nearly 20% of group revenue *Energy-Storage.News, 2026-07 — CATL dossier*. ANALYSIS (High) The lead is narrowing: company-reported ESS share fell from 36.5% to 30.4% across 2025, with Hithium and EVE at roughly 12% each *CATL dossier*. Product spans TENER Stack at 9 MWh on the 587 Ah cell *pv magazine, 2025-05-08 — CATL dossier* and TENER Sodium, the first field-validated sodium-ion BESS at 30+ MWh with 15,000 cycles and a 30-year warranty, which converted to a commercial pipeline within weeks *CATL, 2026-06-22 —*

CATL dossier. It supplied all 19 GWh of Masdar's anchor order *ESS News, 2025-01-20 — CATL dossier*. On the US side it sits on the DoD 1260H list, with Pentagon direct-contracting bars phasing in from June 2026 and DoD procurement barred from October 2027 *CnEVPost, 2026-07-24 — CATL dossier*.

STRATEGY

CATL's celebrated workaround is the License-Royalty-Service model: it supplies patents, process technology, equipment and ramp-up engineers for a fee plus per-unit royalties without owning the plant. Ford's $3.5B BlueOval Battery Park Michigan is the proof point, entering testing in mid-2026 with market-ready cells due before end-2026 *electrive, 2026-06-18 — CATL dossier*, on an LRS book of roughly 139 GWh. ANALYSIS (Moderate) The bind is that LRS is, mechanically, a technology transfer — the precise act MOFCOM's July 2025 catalogue made licensable. Deals already ramping carry forward; each new one is a permission rather than a commercial decision, which is why the reported GM plant for around 2027 should be treated as contingent on Beijing as much as on Detroit. The capital-light US route and the export-control regime are now in direct tension, and the structure that is not — wholly-owned Debrecen, funded by roughly 90% of the HK$41B Hong Kong raise — is European, not American.

3.4 BYD — the cost floor, exporting assembly rather than cells

BYD spans cells, blocks and merchant supply, and Benchmark ranked it the number one global BESS integrator for 2025 at more than 60 GWh and 13% share, overtaking Tesla *Electrek, 2026-05-13 — BYD dossier*. ANALYSIS (Moderate) That crown is scope-dependent — InfoLink still places BYD third in systems — so name the ranking body on every claim *BYD dossier*. HaoHan is the density leader: a 14.5 MWh single DC block on a 2,710 Ah long Blade cell at 233 kWh/m³, roughly 51% above the industry average, with the GC Flux grid-forming PCS in-house and a claimed lifecycle cost below CNY 0.1/kWh *pv magazine, 2025-09-19 — BYD dossier*. It won 11.275 GWh of the Masdar/EWEC round-the-clock project *BYD Energy Storage, 2026-07-09 — BYD dossier*, 12.5 GWh from the Saudi Electricity Company *pv magazine, 2025-02-17 — BYD dossier*, and three repeat Grenergy purchases in Chile. FinDreams also sells cells merchant into Tesla's Shanghai Megafactory *CnEVPost, 2024-06-05 — BYD dossier*. FY2025 group revenue was RMB 803.97B (+3.5%) with net profit down 19% on the domestic price war *CnEVPost, 2026-03-27 — BYD dossier*.

STRATEGY

BYD's cost position comes from consuming its own cells across vehicles and storage, so its localisation problem is narrower than a merchant cell maker's — it needs assembly abroad, not cell technology abroad. The planned line in Brazil, up to BRL 500M, is explicitly module, pack and system integration and not cells *ESS News, 2026-06-22 — BYD dossier*. ANALYSIS (Moderate) That split is the export-control regime showing up in a capital plan: cathode preparation is the controlled step, so cells stay home and the exportable part of the value chain is the enclosure. The second-generation Blade Battery sharpens this — chemistry reports conflict between LMFP and improved LFP *CnEVPost, 2026-03-05 — BYD dossier*, and if it is LMFP its cathode process sits squarely in the restricted catalogue, making any future overseas cell plant a licensing question. ANALYSIS (Moderate) BYD's DC blocks ship without a PCS in some scopes and with GC Flux in others, so its US inverter exposure is a function of scope of supply, not of the company.

3.5 Hithium — the documented case

Hithium is a stationary-only cell and block specialist, ranked top two globally in both total and utility-scale ESS cell shipments for 2025 after third place at 35.1 GWh in 2024, with cumulative shipments past 100 GWh in August 2025 *InfoLink via ESS News, 2026-02-09 — Hithium dossier*. Its differentiation is the large-cell ladder: the 1175 Ah ∞Cell was the first mass-produced kAh-class cell *PR Newswire, 2025-06 — Hithium dossier*, with a 1300 Ah cell in a 6.9 MWh eight-hour-native system targeted for Q4 2026 mass delivery *pv magazine, 2026-06-09 — Hithium dossier*. It runs 10 GWh/yr of module and system assembly at Mesquite, Texas, roughly $200M, first shipments August 2025 *Energy-Storage.News, 2025-08 — Hithium dossier*, and a committed €400M cell-plus-BESS gigafactory in Navarre, Spain with a €81M Spanish grant, targeted 2027 *ESS News, 2026-03-11 — Hithium dossier*. References include 4 GWh of Saudi LDES *PR Newswire, 2025-08-27 — Hithium dossier* and 2 GWh to Ukraine *ESS News, 2026-02-10 — Hithium dossier*. ANALYSIS (Moderate) Financing, not demand, is the constraint — three failed listing attempts against capex-heavy expansion *Hithium dossier*.

STRATEGY

Hithium ran both localisation routes simultaneously and only one survived. The licensing route — a cell-technology licence to Reliance Industries — collapsed in January 2026 when Beijing tightened export controls on battery technology transfer, and Reliance paused its India cell plan; the commercial relationship continues as component supply *Indian Chemical News, 2026-01-12 — Hithium dossier*. The wholly-owned route — Texas assembly and Spanish cells — continues. ANALYSIS (High) That is the whole lesson of this tier in one company: a licence is a technology transfer subject to discretionary permission, while a plant you own transfers nothing, so the capital-heavy path is the only reliably available one. Texas assembly also does real tariff work as US duties on Chinese battery imports step from 7% to 26% in 2026 *ESS News, 2025-06-06 — Hithium dossier*, though assembly does not by itself solve cell-level content rules. Spain in 2027 is where its European content claim becomes real.

3.6 EVE Energy — the challenger being closed out of one market

EVE Energy is the third-ranked global ESS cell maker on InfoLink's FY2025 cut, displaced from second by Hithium, on 71.05 GWh of ESS shipments (+41%) and more than 200 GWh cumulative *InfoLink via ESS News, 2026-02-09 — EVE dossier*. ANALYSIS (Low) Rank claims need care: company messaging still says second, and InfoLink FY2025 is the defensible number *EVE dossier*. It forced the large-cell transition: Mr.Big 628 Ah was the first mass-produced 600 Ah-plus prismatic cell, claiming 50% fewer system components and 50% less installation workload than 314 Ah systems, and the world's first 628 Ah grid-scale station connected at Lingshou, 400 MWh across 80 Mr.Giant units, with more than 30 GWh of 2027 orders in hand *EVE Energy, 2026-02-10 — EVE dossier*. The cycle turned violently: FY2025 revenue of ¥61.47B (+26.4%) produced only 1.44% profit growth, then H1 2026 profit was guided up 95–110% *EnergyTrend, 2026-06-16 — EVE dossier*. At OCP China 2026 it showed a full-matrix 800 V BBU portfolio across rack, distribution and grid layers *EVE Energy, 2026-07-24 — EVE dossier*.

STRATEGY

ANALYSIS (High) The US is closing on EVE from three directions — a reported June 2026 DoD 1260H listing, LG's Tulip-vehicle ITC Section 337 action seeking import exclusion on cylindrical cells, and the broader compliance wall — making minority stakes, licensing and offshore cells the template for how Chinese cell makers serve US demand; direct US exports are already under 4% of revenue on FOB terms *EVE dossier*. Note which of those tools survives the mirror constraint: the 10% stake in the Amplify Cell Technologies JV in Mississippi and Malaysian cells for Tesla are structures that move product and capital, not process; a licence would need a MOFCOM permission. The ITC action targets power-tool cylindrical cells and does not implicate large-format ESS cells, but an exclusion order would establish patent litigation as a second, privately-driven barrier alongside the public one *ESS News, 2026-07-27 — EVE dossier*. ANALYSIS (Moderate) The BBU push has real differentiation and no named wins — 2027-plus optionality, not revenue *EVE dossier*.

3.7 Fluence — a compliant supply chain it cannot yet ship through

Fluence is the Western pure-play integrator, ranked seventh in Wood Mackenzie's first comprehensive ranking *ESS News, 2026-07-14 — Fluence dossier*. ANALYSIS (High) AI data-center demand is a contracted channel rather than pipeline talk: roughly $850M of data-center orders through July 2026 and master supply agreements with two hyperscalers sit inside a record $6.4B backlog *Fluence dossier*. Smartstack replaced the container paradigm with a central skid plus swappable pods, extended to 10 MWh at roughly 680 MWh/acre and up to 40% lower balance-of-plant cost *Fluence, 2026-06-23 — Fluence dossier*, and was selected for Siemens' NVIDIA AI-data-center reference architecture for load smoothing, black start and ride-through in a 136 MW design *Energy-Storage.News, 2026-06-02 — Fluence dossier*. Its US chain runs Tennessee cells, Utah modules, Arizona enclosures and BMS, Texas thermal and South Carolina inverters, with first domestic-content shipments in September 2025 *ESS News, 2025-09-05 — Fluence dossier*. Q3 FY2026 revenue was $649.8M with about $400M delayed by production problems at new contract-manufacturing sites, and FY2026 guidance cut to $2.9–3.1B against record $1.44B order intake *Fluence IR, 2026-08-05 — Fluence dossier*.

STRATEGY

Fluence is the clearest test of whether a regulatory advantage is durable or rented, because it has already been burned from both sides. FY2025 revenue landed at $2.3B with a $68.0M net loss after tariffs paused US projects — the rules moved faster than the supply chain *Fluence dossier*. It then onshored every major component and now states it expects FEOC compliance on the OBBBA (One Big Beautiful Bill Act) deadlines *Energy-Storage.News, 2025-11 — Fluence dossier*. ANALYSIS (Moderate) Execution of that footprint, not demand, is the binding constraint on converting backlog to revenue *Fluence dossier*. The honest reading: the umbrella is real and Fluence is standing under it, but an umbrella is worth nothing to a supplier who cannot ship, and a second domestic cell source is not expected until roughly 2027. The software layer — Mosaic optimising 17 GW and $148M of ARR at FY2025-end — is the part of the moat that no rule change can revoke.

3.8 FlexGen — the layer with almost no policy exposure

FlexGen deliberately left the hardware race and now runs the software and services layer on other vendors' blocks. ANALYSIS (High) Its absence from Wood Mackenzie's AC-block top ten reflects the lane it chose, not competitive failure *FlexGen dossier*. It bought Powin's assets out of Chapter 11 for $36M in August 2025 — hardware and software IP, spare parts, IT systems — lifting its supported base to more than 25 GWh across 200-plus systems in ten-plus countries *Business Wire, 2025-08-19 — FlexGen dossier*, then acquired commissioning firm Clean Energy Services in April 2026 with 125-plus field technicians and 15-plus GWh commissioned *Business Wire, 2026-04-02 — FlexGen dossier*. HybridOS is hardware-agnostic across 65-plus configurations from 22 vendors. Its data-center wedge is BESSUPS with Rosendin: medium-voltage 1–35 kV battery UPS outside the data hall, combining both firms' patents *Business Wire, 2025-06-03 — FlexGen dossier*. European operations launched in June 2026 with projects in the UK, Finland and Sweden *ESS News, 2026-06-22 — FlexGen dossier*.

STRATEGY

FlexGen is worth studying precisely because it is the control case. Software is not a cell, not an inverter and not an imported covered device, so FEOC, tariff and Covered-List exposure all fall away — which demonstrates that policy risk in this tier is a property of the physical thing crossing a border, not of the market being served. ANALYSIS (Moderate) The Powin purchase converted a rival's bankruptcy into a services annuity plus a captive retrofit pipeline, and fleet-under-software is now the metric that matters *FlexGen dossier*. It even sells the policy anxiety directly: a US-engineered BMS software layer sits between foreign-made battery hardware and site communications, addressing the firmware-control concern without replacing hardware. ANALYSIS (Low) Key scale claims date to 2022 company statements and the company is private, so treat share figures as unverified *FlexGen dossier*.

3.9 LG Energy Solution — the sharpest pivot, on rented ground

LG Energy Solution is executing the fastest EV-to-ESS rotation of the Korean cell makers. H1 2026 ESS shipments grew 357% year on year with roughly 86% going to North America *CnEVPost, 2026-08-04 — LGES dossier*, ESS revenue rose about 4.6 times to the high-20% of the group, and the end-2025 ESS backlog stood at 140 GWh *LGES, 2026-01-29 — LGES dossier*. It won Tesla's Megapack 3 LFP cell business at $4.3B, producing at Lansing from 2027 *Electrek, 2026-03-17 — LGES dossier*, and booked more than KRW 3T of new ESS contracts in H1 2026 including AI-data-center deals. Its Vertech arm sells turnkey systems, including 6 GWh / 1.5 GW for DTE supporting the grid that serves the Oracle-managed OpenAI campus at Saline Township *LGES Vertech, 2026-05-27 — LGES dossier*. Q2 2026 operating profit of KRW 113.3B missed consensus of roughly KRW 249B and was a KRW 128B loss excluding US production credits *LGES, 2026-07-30 — LGES dossier*.

STRATEGY

ANALYSIS (High) The durable advantage is regulatory, not technical: FEOC rules and domestic-content credits make its five-site North American ESS network the default non-Chinese supply chain, with the Tesla award and the DTE/Qcells/Excelsior pipeline as direct evidence *LGES dossier*. Two things should temper how much you pay for that. First, the advantage is not yet dominant even in its home market: LGES holds 13.6% of North American ESS supply against CATL's 38.8% and Hithium's 17.2%, so the compliant lane is a premium niche inside a market Chinese suppliers still lead. Second, ANALYSIS (Moderate) profitability quality is the central vulnerability — ex-credit operating losses in most quarters from Q4 2024 through Q2 2026 mean the equity story rests on US production credits persisting long enough for ESS margin to mature *LGES dossier*. Rented advantage plus credit dependence is a levered bet on policy continuity, not a moat.

3.10 Samsung SDI — buying the part of the moat that is not rented

Samsung SDI is small in share and differentiated where AI data centers care. It held 1.4% of global ESS batteries in H1 2026, but 1.6 GWh of those shipments went to AI data centers *CnEVPost, 2026-08-04 — Samsung SDI dossier*, and Q2 2026 delivered its first operating profit in seven quarters at KRW 203.8B, company-attributed to data-center UPS and BBU lines plus ESS *Samsung SDI, 2026-07-30 — Samsung SDI dossier*, with UPS and BBU sales guided to grow more than 70% in 2026. It passed UL's Indoor Large-Scale Fire Test for UPS batteries first in the world, with no propagation and no sprinklers, framed explicitly as a hyperscaler procurement requirement *Samsung SDI, 2026-07-14 — Samsung SDI dossier*. Two billion-dollar-class US ESS deals landed in four months — KRW 1.5T of prismatic supply *Samsung SDI, 2026-03-16 — Samsung SDI dossier* after a more than KRW 2T LFP award delivered as SBB 2.0 from 2027 *Samsung SDI, 2025-12-10 — Samsung SDI dossier* — and US prismatic LFP cell production starts October 2026 at StarPlus.

STRATEGY

ANALYSIS (Moderate) The US LFP conversion is the pivotal execution item: both US deals and the claimed through-2029 order book depend on StarPlus landing on time, and the roughly KRW 1.6T L&F cathode agreement shows the compliant chain being built deliberately *Samsung SDI dossier*. That cathode deal is the most instructive move in this chapter. Cathode is simultaneously where FEOC content rules bite and where MOFCOM's controls apply, so a Korean cathode source buys immunity from both regimes at once — compliance and supply security in a single contract. ANALYSIS (Moderate) Safety certification is the deliberate differentiation against Chinese price leaders, targeting the one dimension hyperscalers will pay a premium for *Samsung SDI dossier* — and unlike a content rule, a fire-test result is a physical property that no rule change can revoke. ANALYSIS (Moderate) Profit quality is the watch item: the Q2 print included tariff refunds and credits are load-bearing after seven loss quarters *Samsung SDI dossier*.

3.11 Panasonic — the incumbent of the rack, not the grid

Panasonic is a cell maker whose data-center exposure sits at the rack layer rather than in grid storage; it appears in no BESS integrator ranking. It claims roughly 80% of the data-center distributed-power market and has published a ¥800B FY2029 data-center storage target that its CEO calls a minimum commitment *Panasonic, 2026-03-25 — Panasonic dossier*. The ramp is visible in the numbers: Q1 fiscal 2027 data-center storage sales were ¥113B, 1.9 times a year earlier, against a ¥550B FY2027 forecast, in a quarter that produced the best Q1 operating profit in 41 years *Panasonic, 2026-07-30 — Panasonic dossier*. Behind it sits ¥350B of dedicated capex across FY2026–28, Kansas De Soto EV lines physically repurposed to data-center cells, and Mexico module sites mass-producing from FY2027 and FY2028 *ESS-News, 2026-06-08 — Panasonic dossier*. Next-generation capacitor backup units built with Panasonic Industry map onto NVIDIA's rack-side supercapacitor tier.

STRATEGY

ANALYSIS (High) The pivot is structural rather than a hedge — dedicated capex, EV assets redeployed rather than idled, and Japanese cell capacity tripling by FY2029 *Panasonic dossier*. On policy, Panasonic is the cleanest beneficiary of how the Covered List is drawn: a BBU is a DC battery module and a server PSU is a unidirectional AC-DC converter, so neither meets the bidirectional prong on a plain reading, regardless of where it is made. ANALYSIS (Moderate) Its exposure is commercial, not regulatory. The 80% share claim is self-reported with no third-party verification, and both Samsung SDI's challenge and reported industry-wide BBU cell shortages suggest the moat will be contested as the market grows *Panasonic dossier* — indeed both Panasonic and Samsung SDI are reported supply-short in BBU cells *The Elec, 2026-07 — Samsung SDI dossier*, which is a capacity story, not a share story. US data-center battery production does not start until fiscal 2028.

3.12 Jinko — a module giant using storage as a lifeboat

Jinko is the world's largest module shipper at 86 GW in 2025, its seventh consecutive year and tied with LONGi *InfoLink, 2026-02 — Jinko dossier*, and it took its first annual loss in twelve years: FY2025 revenue of RMB 65.5B (−29%) and a RMB 4.45B net loss *JinkoSolar, 2026-04-16 — Jinko dossier*. Gross margin recovered to 8.3% in Q1 2026 from 2.2% for FY2025 as the loss narrowed to RMB 463.5M *JinkoSolar, 2026-04-29 — Jinko dossier*. Storage is management's stated profitability lever: 5.2 GWh shipped in 2025 against only about 1.7 GWh revenue-recognised, guided to more than double in 2026, with more than 10 GWh of signed and high-probability orders. SunTera G3 delivers 6.25 MWh on 587 Ah cells at above 95% round-trip efficiency with grid-forming modes including black start *ESS-News, 2025-04-14 — Jinko dossier*. In August 2026 it launched Sunny 365, whose AIDC edition pairs 680 W Tiger Neo modules with SunTera storage and claims 800 V HVDC compatibility *Jinko ESS, 2026-08-04 — Jinko dossier*, alongside a planned 1 GW data center near Zhongwei, Ningxia *GuruFocus, 2026-06 — Jinko dossier*.

STRATEGY

Jinko shows policy exposure resolving at the product line rather than the company. Its module business is effectively priced out of the US by Jinko-specific AD/CVD (antidumping and countervailing duty) rates of up to about 246% on Vietnamese output, and the Tiger Neo 3.0 rollout excludes the US outright *pv magazine USA, 2025-04-*

21 — Jinko dossier. Its storage business is a block-and-system business whose US route would require US assembly it has not announced. ANALYSIS (High) Storage here is survival, not diversification: with FY2025 module gross margin at 2.2% and the whole Chinese chain loss-making, the ESS ramp is the stated primary profitability lever *Jinko dossier*. ANALYSIS (Moderate) The AIDC moves are a demand-side hedge — owning or serving data-center load monetises its own output domestically while trade barriers cap the western module business — structurally logical and unproven *Jinko dossier*. Note the gap between shipped and revenue-recognised GWh: this is an order book, not yet an earnings stream.

3.13 Delta Electronics — the incumbent standard inside the rack

Delta is the volume leader in AI server power: number one global power-supply vendor, reported at more than half of AI-server AC/DC supply with brokerage estimates above 60%, and roughly 70% of Blackwell-platform PSUs *WAWT, 2026-04-20; KGI, 2025-07-01 — Delta dossier*. AI data-center power plus cooling crossed 50% of revenue for the first time in Q2 2026, on record revenue of NT$183.2B (+48%) and record EPS of NT$9.68, with 2026 capex raised 50% to NT$70B *Delta Q2 2026, 2026-07-29 — Delta dossier*. Its 800 VDC stack is the deepest published: a 660 kW in-row power rack built from six 110 kW shelves each embedding an 80 kW BBU — 480 kW of BBU per rack — on 18.5 kW PSUs at up to 98%, an SST converting medium-voltage AC to 800 VDC at 98.5%, a silicon-carbide e-fuse cutting in under 3 µs, and a claimed grid-to-chip efficiency gain above 4% to roughly 92% *Delta 800 VDC deep-dive, 2026-05-28 — Delta dossier*. Ramp guidance: ±400 VDC mass production Q3 2026, 800 VDC small volume Q4 2026, true volume 2027 *TrendForce, 2026-06-15 — Delta dossier*.

STRATEGY

ANALYSIS (High) Delta is structurally advantaged into the 800 V generation — first-qualified across GB200 and GB300 shelf sets, a reported NVIDIA DC-DC exclusive, the deepest published stack, and a reported $2B silicon-carbide supply lock with Infineon *Delta dossier*. Its policy position is the chapter's key teaching case. Delta is Taiwanese and manufactures across Asia, so it is foreign-produced under the Covered List's origin test — nationality does not save it. But its rack-power content is unidirectional AC-DC with wired management, which fails the bidirectional prong on a plain reading, while its storage systems and EV-charging equipment do not. ANALYSIS (Moderate) The exposure therefore sits in the smaller, non-rack parts of the portfolio, and this is a read of the rule's structure, not a ruling on Delta. ANALYSIS (High) The margin narrative has peaked near-term: management guided H2 2026 gross margin no higher than about 35.6% against Q1's record 37.0%, with memory and MOSFET cost inflation and a shortened depreciation life *Delta dossier*.

3.14 LITEON — the BBU franchise becoming mandatory content

LITEON is the second source in AI server power at roughly 30% against Delta's 60%-plus on brokerage estimates, and its structural asset is battery backup. BBU shipments in 2025 ran about 30 times the prior year, and BBUs move from optional on GB200 to standard on GB300 and Rubin *LITEON dossier; fiisual*. Its shelves are specific: a 12 kW 2U 50 V BBU shelf giving 45 seconds of full-power backup, a 22 kW 3U 800 V supercapacitor shelf giving 60 seconds, and designs targeting two-minute ride-through so coolant distribution units keep running, with microsecond switching from PSU-BBU firmware co-design. Its GB300 NVL72 collaboration with NVIDIA cut grid peak load by about 30% *OCP 2025, 2025-10-13 — LITEON dossier*. Q2 2026 set all-time records: revenue NT$52.7B (+30%), EPS NT$3.14 (+126%), gross margin 27.2%, with cloud-related revenue up more than 100% *LITEON Q2 2026, 2026-07-31 — LITEON dossier*. It committed $919M to a McKinney, Texas site of more than 650,000 sq ft focused on HVDC power racks, taking non-China capacity to 60% by end-2026 and 70% by 2027 *LITEON, 2026-07-13 — LITEON dossier*.

STRATEGY

ANALYSIS (High) The architecture shift amplifies LITEON's strongest product — the 800 V rack embeds BBU and supercapacitor shelves, so what was optional becomes non-discretionary content *LITEON dossier*. ANALYSIS (High) It is a step behind Delta on the GPU-side ramp and is routing around that through ASIC customers: mass production Q1 2027 after a 13-week qualification from August 2026 sampling, ASIC clients first, GPU clients potentially Q2 2027 *LITEON dossier*. Read McKinney as optionality on a rule that does not yet cover its products: like Delta, LITEON is foreign-produced but its unidirectional rack power falls outside the two-prong test on a plain reading, so US manufacturing is a tariff and proximity hedge today that would become a compliance asset only if an origin test spread beyond inverters ANALYSIS (Moderate). ANALYSIS (Moderate) The balance sheet is carrying the ramp visibly — short-term loans up 132% to NT$31.4B and a 200M-share placement authorisation *LITEON dossier*.

3.15 Megmeet — the mainland challenger whose exposure is procurement, not product

Megmeet is the only mainland-China power supplier NVIDIA has named for GB200 NVL72 power and a member of its 800 V HVDC ecosystem *SmBom, 2025-05-22 — Megmeet dossier*. It has shown the full grid-to-GPU chain: an AC compensator, backup and BBU shelves, an SST above 98.5% efficiency outputting 800 VDC, a 1 MW-plus sidecar, 33 kW and 110 kW power shelves, and point-of-load bricks. The Kyber sidecar claims 5% better end-to-end efficiency and 45% less copper *Megmeet, 2025-05-21 — Megmeet dossier*, and a 110 kW liquid-cooled shelf followed for MGX in May 2026. It opened a 360 kW AI data-center power test lab in Dallas with a roadmap to 1.5 MW *Megmeet, 2026-06-26 — Megmeet dossier*. The cost has been brutal: FY2025 revenue of ¥9.40B (+15.1%) produced net profit of ¥145.8M (−66.6%), just ¥25.8M excluding non-recurring items (−93.0%), and the first negative operating cash flow year at −¥139M *SZSE filing, 2026-04-29 — Megmeet dossier*. The company itself told 21jingji in November 2025 that AI-server PSU orders were still small-batch and small-value with uncertain timing *21jingji, 2025-11-04 — Megmeet dossier*.

STRATEGY

ANALYSIS (Moderate) Megmeet's wedge into a Delta-dominated market is architecture timing rather than price — first mover among mainland vendors on the sidecar and SST full chain *Megmeet dossier*, and its geographic build-out across Thailand, India, Germany and the Dallas lab is a wholly-owned footprint with no licensing structure to be denied. Its bind is the inverse of Sungrow's. Its core rack products are unidirectional PSUs, shelves and an SST, none of which meets the bidirectional prong on a plain reading, while its storage backup shelves and EV-charging modules do ANALYSIS (Moderate). So the rule that captures Chinese inverter makers largely misses Megmeet's growth products — but nationality-based procurement screening, which is a different mechanism entirely, still stands between it and US hyperscaler sockets. ANALYSIS (High) Expectations are front-loaded relative to disclosure: the company said AI orders were immaterial through 2025 while consensus embeds roughly a six-fold profit rebound, and the H1 report due 27 August 2026 is the first hard test *Megmeet dossier*.

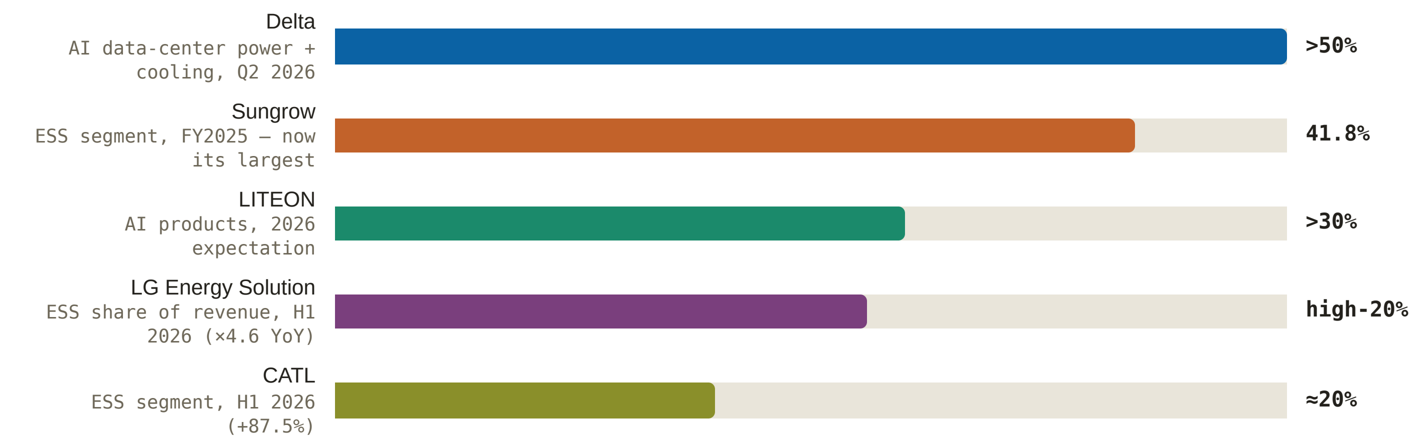
3.16 Sinexcel — one company, two opposite exposures

Sinexcel sells modular power conversion to other people's battery systems — the merchant PCS lane beneath the integrator layer. Its StellaON 1250K/1575K grid-forming flagship runs at 690 Vac with 99% peak efficiency and took more than 500 MW of orders in H1 2026 ahead of its formal European launch *Sinexcel, 2026-07-07 — Sinexcel dossier*, having been among the first PCS to win Poland grid certification *Energy-Storage.News, 2026-04 — Sinexcel dossier* and entered Japan's frequency-regulation market on a 2 MW/8 MWh project *Sinexcel, 2026-07-13 — Sinexcel dossier*. It was named a BloombergNEF Tier 1 inverter manufacturer in Q2 2026. FY2025 revenue was RMB 3.463B (+14.1%) with net profit of RMB 476M (+11.0%); overseas storage revenue of RMB 645M grew 50.1% at a 61.3% gross margin while the domestic storage line fell 3.7% *东方财富财富号, 2026-06-24 — Sinexcel dossier*. Its AIDC division, formed June 2025, is developing HVDC systems and SSTs; current shipping exposure is a 36 kW HVDC rectifier module, third-party-reported as supplied to Vertiv, with own-brand HVDC pre-revenue as of February 2026 *海之理财, 2025-10-09 — Sinexcel dossier*.

STRATEGY

Sinexcel is the cleanest demonstration that the Covered List is a product test, not a company test. Its flagship is a bidirectional storage PCS with remote monitoring — both prongs met — and it manufactures in Huizhou, Suzhou, Malaysia and California; whether the Californian output clears the domestic component-cost threshold is not disclosed anywhere in the corpus, so its US position on that product is genuinely unknown. Its 36 kW HVDC rectifier module is unidirectional and falls outside on a plain reading, which is precisely why the AIDC line is the one with an unobstructed US route, through an established infrastructure vendor rather than direct ANALYSIS (Moderate). ANALYSIS (High) Overseas utility PCS is the margin engine — 61.3% gross margin on exports versus commoditised domestic pricing, with grid certifications functioning as queue-jumps into tender shortlists *Sinexcel dossier*. ANALYSIS (Moderate) The AIDC line is optionality and pre-revenue; the H1 2026 report due 11 August 2026 is the first disclosure checkpoint *Sinexcel dossier*.

Figure 16: How completely each layer has converted — share of company revenue from the AI-data-center or storage line (%, latest disclosed period)



Source: Delta Q2 2026, 2026-07-29 — Delta dossier; EnergyTrend, 2026-04-01 — Sungrow dossier; LITEON Q2 2026, 2026-07-31 — LITEON dossier; LGES, 2026-07-30 — LGES dossier; Energy-Storage.News, 2026-07 — CATL dossier. Note: periods and segment definitions differ by company and are not strictly comparable; each bar is direct-labelled with the figure as disclosed. Bar widths are scaled to the largest value shown.

THE SIXTEEN, SORTED THE WAY THAT MATTERS

COMPANY	LAYER	MANUFACTURING FOOTPRINT	POLICY EXPOSURE	STRATEGY IN ONE LINE
Sungrow	AC-system integrator (+PV inverters)	China	Highest — bidirectional PCS meets both FCC prongs; EU phase-out by Apr 2027; little cathode-control exposure	Ride both Wood Mackenzie crowns and grid-forming maturity into Europe and the Gulf while the US door narrows
Tesla	AC-system integrator (+own cell)	US (Lathrop, Brookshire, Sparks), Shanghai	Low — buys compliance via Korean cells and onshoring	Standardise the unit, integrate at site level in Megablock, monetise the fleet in software

COMPANY	LAYER	MANUFACTURING FOOTPRINT	POLICY EXPOSURE	STRATEGY IN ONE LINE
CATL	Cell + DC block	China; Hungary (Debrecen); US via licence only	1260H procurement bars; LRS licensing now needs MOFCOM permission	Defend the cell crown, sell the sodium transition, and localise where ownership — not licensing — is possible
BYD	Cell + DC block + merchant cell	China; planned Brazil assembly	Cells stay home under cathode controls; block exposure depends on scope of supply	Export assembly, not cells — let EV-scale Blade economics set the price floor
Hithium	Cell + DC block (stationary only)	China; Mesquite TX assembly; Navarre Spain cells 2027	Tariffs 7%→26%; licensing route already closed by export controls	Own the plants, climb the cell ladder to 1300 Ah, and sell eight-hour duration where mandates exist
EVE Energy	Cell (+emerging BBU)	China; Malaysia; Hungary 2027; 10% of a US JV	Reported 1260H; ITC cylindrical exclusion risk; US exports already under 4% of revenue	Force the 600 Ah-plus transition, serve the US through minority stakes and offshore cells
Fluence	AC-system integrator (+software)	US: TN cells, UT modules, AZ, TX, SC	Beneficiary — domestic content and FEOC compliance are the product	Convert a compliant supply chain and two hyperscaler agreements into shipped revenue
FlexGen	Software and services on others' hardware	US (no hardware manufacturing)	Minimal — nothing covered crosses a border	Own the controls and service annuity on any vendor's fleet, including orphaned ones
LG Energy Solution	Cell + integrator (Vertech)	Five North American ESS sites; Korea; Poland	Beneficiary — the default non-Chinese US chain, but credit-dependent	Convert EV overcapacity into an ESS franchise faster than anyone, on US credits
Samsung SDI	Cell (ESS, UPS, BBU)	Korea; StarPlus Indiana LFP from Oct 2026	Beneficiary — non-FEOC chain secured down to Korean cathode	Sell certified safety and non-Chinese prismatic cells at a premium into AI data centers
Panasonic	Cell + rack power (BBU, CBU)	Japan; Nevada; Kansas; Mexico modules	Outside the FCC prongs on a plain reading; exposure is commercial	Redeploy EV capacity into rack backup and defend an unverified 80% share as the market grows
Jinko	Modules + DC block/system challenger	China; Southeast Asia; Jacksonville FL; Middle East	Modules already excluded by AD/CVD to ~246%; storage lacks a US assembly route	Use storage — and now data-center load itself — to escape a loss-making module market

COMPANY	LAYER	MANUFACTURING FOOTPRINT	POLICY EXPOSURE	STRATEGY IN ONE LINE
Delta	Rack-power ODM (+facility)	Taiwan and Asia; Thailand	Foreign-produced, but unidirectional rack power falls outside; PCS and EV charging do not	Hold the incumbent standard inside the NVIDIA rack from SST down to DC-DC brick
LITEON	Rack-power ODM	Asia; McKinney TX; non-China to 70% by 2027	Same read as Delta; Texas is optionality on a rule that has not arrived	Turn the BBU franchise into mandatory rack content and reach GPU sockets via ASIC customers first
Megmeet	Rack-power ODM	China; Thailand; India; Germany; Dallas lab	Rack products largely outside the FCC prongs; exposure is procurement screening	Buy early architecture position in 800 VDC with incumbent-funded losses, and prove revenue on 27 Aug
Sinexcel	Merchant PCS + rack-power module	Huizhou, Suzhou, Malaysia, California	Split — storage PCS meets both prongs; the HVDC module does not	Earn on high-margin overseas grid-forming PCS while the AIDC module rides in through a partner

Every figure behind this table is cited in sections 3.1–12.16. Layer assignments and policy-exposure reads are analysis, not sourced rulings; FCC and FEOC mechanics are in the policy chapter.

3.17 What this means for your capital

Three conclusions follow from sorting this tier by layer rather than by flag.

First, the layer that has converted most completely to this market is rack power, not grid storage. Delta crossed 50% of revenue from AI data-center power and cooling in a single quarter; no storage name in the file is close on that measure. That matters because rack content is specified inside NVIDIA reference designs on a published clock — ±400 VDC in volume from Q3 2026, true 800 VDC volume in 2027 — and because the architecture makes BBU and supercapacitor content non-discretionary rather than optional. Where a grid-storage order is won on price per kWh against a dozen bidders, a rack-power socket is won once, at qualification, and then held for a product generation. The corollary is concentration risk: these companies now rise and fall with one customer's rack cadence.

Second, the compliant lane's advantage is real, priced, and rented. LG Energy Solution, Samsung SDI, Panasonic, Fluence and Tesla are all monetising the same thing — being able to sell into the US without a provenance argument. But look at what each one actually owns. Fluence owns a physical chain it is still learning to run, and its FY2026 guidance cut proves an umbrella is worthless to a supplier who cannot ship. LG Energy Solution owns North American capacity but posted ex-credit operating losses in most quarters from Q4 2024 through Q2 2026, so its returns are a levered position on production credits persisting. Samsung SDI owns two things a rule change cannot revoke — a Korean cathode source and a world-first UL fire-test result — which is why its premium is the best-founded in the group. Panasonic's rack products sit outside the inverter rule's structure entirely, so its risk is capacity and a contested share claim, not policy. Rank the rented advantages by what remains if the rule moves, and the ordering changes materially.

Third, the Chinese leaders are not being pushed out; they are being pushed into a more capital-intensive shape. They still lead the market they are supposedly excluded from — CATL holds 38.8% of North American ESS supply against LG Energy Solution's 13.6%. What changed is the route. The cheap route was licensing: transfer process technology, collect a royalty, spend nothing. That route now requires a Chinese permission, and the two documented attempts to use it — Hithium with Reliance, and Amara Raja's with Gotion — both ended. What is left is owning the plant: Hithium in Mesquite and Navarre, BYD's assembly line in Brazil, CATL's wholly-owned Debrecen. Own-plant localisation is slower, absorbs far more capital, and shows up as depreciation before it shows up as share — so expect the Chinese names' reported margins to compress precisely as their compliant capacity arrives. That is the trade they are making, and it is the right one, because it is the only one available.

What to monitor, on a calendar. Near-dated disclosure first: Sinexcel's H1 on 11 August 2026 for any AIDC revenue, EVE around 20 August, Megmeet's H1 on 27 August as the test of whether its 800 VDC position is revenue, Jinko around 27 August, and Sungrow and BYD on 29 August. Then execution: Samsung SDI's StarPlus US LFP start in October 2026, Hithium's 1300 Ah mass delivery in Q4 2026, Tesla's Megapack 3 customer deliveries in H2 2026, LITEON's 800 V mass production in Q1 2027, Delta's true 800 V volume and LG Energy Solution's Lansing cells in 2027, Hithium's Navarre plant in 2027, and Panasonic's US data-center battery production in fiscal 2028. Then policy: the DoD procurement bar on CATL from October 2027, FCC conditional-approval applications closing 1 January 2028, and the Buy American domestic-content threshold stepping from 65% to 75% in 2029.

WHAT WOULD CHANGE THIS VIEW

- The FCC conditional-approval path starts granting at scale. Applications run to 1 January 2028 with no published review timeline. If approvals become routine, the new-authorization freeze stops functioning as a pricing umbrella and the compliant lane's premium deflates — this is the single largest falsifier of the compliant-lane thesis.
- MOFCOM begins routinely licensing cathode technology. If licensing reopens, Chinese cell makers regain a capital-light localisation route, own-plant depreciation stops being the price of US access, and the Korean and Japanese premium compresses quickly. Watch for any new LRS-style deal actually closing after the July 2025 catalogue.
- The US ramps land, or slip again. Fluence's new contract-manufacturing sites reaching production early in FY2027 and StarPlus starting in October 2026 would convert the compliant lane from a story into shipped revenue. A second slip at either would show the umbrella exists but nobody can trade under it.
- Megmeet's 27 August H1 shows material AI-DC revenue. That would make rack power a genuine three-way contest and mean Delta's share assumptions — and the premium paid for them — are too high. A weak print validates the challenger discount.
- BBU and supercapacitor supply-shortness resolves. Both Panasonic and Samsung SDI are reported short on BBU cells. Ample supply removes the scarcity component from rack-storage economics and turns Panasonic's claimed share into a price-competition question.
- 800 VDC slips past 2027. Every qualification date in this chapter — LITEON's Q1 2027, Delta's volume ramp, Megmeet's sidecar position — is anchored to that clock. A generational delay resets the whole vendor-list-reset argument and hands time back to the incumbents.

This chapter is analysis for understanding commercial position; it is not a recommendation to buy or sell any security.

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Typeset source: repository-information/aidc-coverage-universe-print.html

Rebuild every edition with: node scripts/build-aidc-report-pdf.mjs

Developed by: ShadowAISolutions